Technical supplement to *Noise Covariance Defines Reconstruction Geometry: Optimal Filtering, Weighted Subspaces, and Tied Linear Autoencoders*

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This supplement carries the derivations, proofs, implementation details, verification grids and per-file diagnostics that the main paper cites as "Supplement, Section Sn". It is self-contained on those points and assumes the notation of the main paper's Section 2. Section numbers below are the targets of those citations.

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S1 How the main paper's citations map into this supplement

The main paper cites this document by the labels below. Each entry names what is here and, where relevant, what the extended arXiv preprint calls the same material, so a reader holding either document can move between them.

Evidence and reproduction. Every quantity in the main paper and in this supplement is read from a file in the versioned evidence root `evidence/`, whose complete SHA-256 manifest is checked by `python3scripts/verify_evidence.py--require-complete`. The figures and generated tables are rebuilt by `figures_src/make_paper1_evidence.py` and `figures_src/make_part2_figures.py`, which read every value from a registered file and hard-code none; `generated/SOURCE_MAP.md` and `generated/SOURCE_MAP_part2.md` map each output cell to its source file and field. The claim-to-evidence table (`CLAIM_EVIDENCE_TABLE.md`) lists, per claim, its mathematical assumption, dataset and split, output field, code revision, output path, figure or table, and limitation; its paths, revision forms and a set of concrete values are machine-checked by `scripts/validate_claim_table.py`.

Table S1: Citation map. "Preprint" names the corresponding section of the extended arXiv preprint, which carries the same material as its appendices.

cited as	here	content
S1	§S2	Provenance of every statement: established, specialized, corollary, synthesis, or derived here; related work
S2	§S4	Proof of the fixed affine-estimator risk identity; model-order correction for maps of unequal trace
S3	§S12	Pullback of the noise metric; excited-support proposition and its proof
S4	§S10	Proof of the Bridge Theorem, including the minimum value and the rank-deficient completion
S5	§S8	Untied encoder: what changes and what does not
S6	§S12	Architecture-neutral loss selection: the isometry axiom and its converse
S7	§S12	Global relative-covariance bound, and the proof of the first-order displacement proposition
S8	§S10	Restricted (Kronecker) bridge and the rank-one NFPA corollary
S9	§S12	Proof of the exact risk decomposition and of the restriction-cost lower bound
S10	§S6	Time-shift and joint-channel optimal filters
S11	§S8	Tied versus untied autoencoders at a fixed budget
S12	§S8	Alternating least squares for the separable class: updates, gauge, cost
S13	§S14	The nine frozen regression gates (S1–S9) in full
S14	§S14	Covariance-perturbation grids: all arms, all seeds
S15	§S14	Structural-prior grids and restart diagnostics
S16	§S14	Excitation study: per-cell recoveries and the post-hoc reading
S17	§S20	Archive identifiers, run commits, manifest digests, configuration inventory

S2 Provenance of the statements

S3 Related Work

The equivalences assembled here each have an established lineage, and covariance-aware low-rank approximation itself is not new. This section places the paper against the nearest prior work, marks each statement of Sections 2–2.2 as established, specialized, a corollary, a synthesis, or derived here (Table S2), and states what the paper adds beyond the placement.

Inverse-covariance weighting and matched filtering. Generalized least squares dates to Aitken [1], and the matched or optimal filter is its signal-processing form [2–4]. In cryogenic calorimetry the same construction is the standard amplitude estimator [5–7]. This literature fixes the weighting but assumes a known one-dimensional template; it does not ask what happens when the signal model itself is learned.

Weighted and probabilistic low-rank models. Weighted low-rank approximation with a general quadratic weight goes back to Gabriel and Zamir [8], and its alternating solution and the distinction between element-wise weights (no closed form) and a full row or column metric (a closed form after transformation) are classical [9]. Probabilistic PCA assumes isotropic noise [10], factor analysis allows a diagonal covariance [11, 12], and weighted or expectation–maximization PCA accommodates nonuniform weights [13, 14], with the underlying EM machinery from Dempster et al. [15] and convergence results for block-coordinate schemes from Tseng [16]. The symmetric-definite generalized eigenproblem $\hat{R}_x v = \lambda \Sigma v$ and orthogonal iteration for it are textbook material [17, Ch. 8]. This line is usually presented in terms of weights rather than of a stated measurement model, and its rank-1 case is not identified there with the detector-physics optimal filter; those are matters of presentation, and nothing in Section 4.2 is new as an estimator.

Covariance-aware and multilinear matrix decompositions. Both learned classes of this paper sit in an established family, and the precise optimization problems should be compared. Allen et al. minimize $\|X - UDV^T\|_{Q,R}^2 = \operatorname{tr}\big(Q(X - UDV^T)R(X - UDV^T)^T\big)$ over factors orthonormal in the Q- and R-inner products, with fixed row and column precision operators, and reduce the positive-definite case to an ordinary SVD of $Q^{1/2}XR^{1/2}$ [18]. The Bridge Theorem is the one-sided case $Q = \Sigma^{-1}$, $R = I$ of that problem, written in encoder–decoder form; Theorem 3.1 is therefore a specialization of a known result, and what Section 4.3 adds is the tied-adjoint reading, the explicit gap and rank-deficient cases, and the untied statement of Appendix S13.2. The separable class is different from

the generalized-least-squares decomposition rather than a case of it: the GMD has an unrestricted rank-k factor pair under a two-sided norm, whereas NFPA has a *product* subspace $\text{range}(U_t) \otimes \text{range}(U_c)$ of rank $k_c k_t$ under a Kronecker-separable covariance — the Tucker-2 structure of multilinear PCA [19, 20] applied to per-axis whitened data, close in spirit to matrix-normal PCA [21]. The Kronecker-restricted bridge (Proposition 4.4) is not a new estimator in this family; what it adds is the placement — the same Σ^{-1} metric, the same tied adjoint encoder, and the same degeneracy to the optimal filter at rank one, with the restriction stated as a change of admissible class rather than of objective — so that the cost of the prior and the effect of the metric can be measured separately at matched rank.

Positional encodings and the noise precision. Laplacian-eigenvector positional encodings for attention on graphs [22], and their reading of the sinusoidal encodings of Vaswani et al. [23] as the cyclic-grid case, are summarized by Bronstein et al. [24, §5.4]. Proposition 4.1 is the observation that, for periodic stationary detector noise, the eigenbasis of the precision operator is the same Fourier basis with the inverse-PSD weights of the optimal filter attached, which supplies a measured rather than assumed operator for that construction; the design suggestions it invites are recorded as untested in Section 7, and no empirical claim about attention models is made.

Linear autoencoders and PCA. That linear autoencoders under Euclidean loss recover the PCA subspace is classical [25, 26], with the loss landscape and the role of weight tying clarified more recently [27, 28]. These autoencoder results are stated in the Euclidean metric; the covariance-weighted optimum itself is the weighted low-rank approximation of the previous paragraphs. The step taken here is to impose the constraint $P^\dagger \Sigma^{-1} P = I_k$ and a covariance-weighted loss in the encoder–decoder parameterization, so that the tied Σ^{-1}-adjoint encoder appears as an adjointness condition rather than an architectural convenience. The subspace geometry itself is standard [29, 30], as is the perturbation theory used in Remark 3.4 [31–33].

Noise-aware learned reconstruction in physics. Learned denoisers are increasingly applied to detector traces. The Xenon-TPC study of Parker et al. [34] trains its unsupervised model under a covariance-dependent Stein unbiased risk estimate, whose divergence term is weighted by the noise covariance; that is an estimate of a *clean-signal* risk, not a residual likelihood, and it is the nearest published instance of the distinction drawn in Section 2.2. Related strands train under structured noise [35, 36], predict heteroskedastic uncertainty [37], or embed physical constraints in the objective [38, 39]; geometric deep learning supplies the language for architecture priors [24], and latent-manifold treatments

of event reconstruction are developing separately [40]. The present paper argues that, for a quadratic residual loss, the residual metric is not a free choice once a covariance model has been asserted — isotropic MSE on the residual asserts $\Sigma = \sigma^2 I$ — while a supervised clean-target loss is a different and legitimate choice of risk (Section 2.2).

Learned geometry versus measured geometry. A separate line of detector machine learning learns the geometry rather than measuring it. GravNet and GarNet map every sensor to coordinates in a learned space and weight messages between sensors by a distance potential in that space [41]; object condensation constructs a potential-based loss whose minimizer places one representative per physical object and attaches to each vertex a learned weight expressing whether it can represent its object [42]; and cross-detector pre-training recipes such as Panda Diplomacy learn one sensor-level representation for several imaging detectors from the data alone [43]. These are the natural tools when the geometry of the measurement is irregular or unknown. The present paper treats the complementary case in which the geometry *is* known, because a calibration run measures it: the inner product between readout samples is Σ^{-1}, its eigenbasis under stationary noise is the Fourier basis with the inverse-PSD weights attached (Proposition 4.1), and the loss is then not a design choice but the unique *quadratic* residual functional invariant under the noise isometries (Proposition 3.3, Appendix S13.1). Where object condensation constructs a loss and shows that its minimizer has the wanted structure, the argument here runs the other way: the noise model fixes the loss, and the question becomes which admissible signal classes—template, subspace, product subspace—the fixed metric admits. A learned coordinate space applied to a calibrated readout must rediscover Σ^{-1} or pay for the difference in the likelihood; Section 5 measures that difference at matched rank.

What this paper adds. Table S2 marks the status of each formal statement. Placing recalled identities in one metric is not by itself a contribution, and the paper does not rest on it. What is added is, first, the risk distinction of Section 2.2: the classical unbiased-risk identity, specialized to fixed linear reconstructions in whitened coordinates, that says exactly which clean-signal risk the weighted residual predicts, the counterexample showing that the Euclidean clean-signal risk can order the other way, and the Davis–Kahan corollary in relative covariance error (Remark 3.4); second, the excited-support observation and the pullback form (Propositions 2.4 and 2.4), which carry the metric to the learned representation and to the training distribution; third, the finite-sample and rank-deficient bookkeeping of the degeneracies (Theorems 4.2 and 3.1); and fourth, an empirical programme that separates the three questions these identities invite — whether a larger or restricted signal model reconstructs better in the assumed geometry, whether the assumed covariance calibrates on the target data, and whether the preferred reconstruction is closer

to a recorded truth in a named metric — and reports its failures. The last question requires a release with an injected reference channel, which is why the TIDMAD stage exists.

Table S2: Status of the formal statements against the nearest prior work. "Established" means the result is standard and is restated for the hierarchy; "specialized" or "corollary" means a known result is written for the one-sided precision Σ^{-1}, the whitened data, or this geometry; "synthesis" means a known argument assembled as the axiom it is; "derived here" means an elementary observation proved in this paper for which no priority is claimed. No statement in the table is claimed as an independently substantiated novelty; the contribution of the paper lies in the unified interpretation, the explicit assumptions and limitations, and the evidence that separates numerical equivalence, calibration and scientific recovery.

Statement	Status	Nearest prior work
OF as GLS amplitude (§4.1)	established	Aitken; matched filtering [1, 2, 5]
Rank-1 population equivalence (Thm. 4.2)	established / specialized	spiked-covariance leading eigenvector [10]
Exact degeneracy to OF (Thm. 4.2)	derived here, elementary	—
Bridge Theorem (Thm. 3.1)	specialized	weighted low-rank approximation, GMD with $Q = \Sigma^{-1}$, $R = I$ [8, 9, 18]; linear AE = PCA [25]
Untied optimum (Prop. S13.3)	specialized	Eckart–Young in whitened coordinates
Architecture-neutral metric (Prop. 3.3)	synthesis (axiom)	Fisher metric of the Gaussian family [44]
Kronecker-restricted bridge (Prop. 4.4)	specialized	MPCA / Tucker-2, matrix-normal PCA [19–21]
Stationary precision eigenbasis (Prop. 4.1)	established	circulant diagonalization [45]; Laplacian PE [22]
Fixed-map risk identity (Prop. 2.1)	established / specialized	unbiased risk estimation [46, 47]; covariance-weighted SURE in detector denoising [34]
Relative covariance-error bound (Rem. 3.4)	corollary	Davis–Kahan [31, 48]
First-order displacement under covariance error (Prop. 3.2)	specialized	invariant-subspace perturbation [49], applied to $A(I + E)^{-1}$
Risk decomposition and restriction-cost bound (Prop. 3.3)	derived here, elementary	approximation–estimation split; Ky Fan maximum principle
Pullback of the noise metric (Prop. 2.4)	established	Fisher information under reparameterization [44]
Excited support (Prop. 2.4)	derived here, elementary	—

S4 Optimal-filter derivations and the risk identity

S5 Detailed Optimal-Filter Derivations and the Cramér–Rao Bound

This appendix expands the summary statements of Section 4.1: unbiasedness, variance, Fisher information, and the energy-resolution formula for the OF estimator.

Convention. The primary convention is a real time-domain trace $x = As_0 + n \in \mathbb{R}^d$ with a real amplitude A and real Gaussian noise $n \sim \mathcal{N}(0, \Sigma)$, Σ circulant under the periodic stationary model of Appendix S17, $\Sigma = F^\dagger \operatorname{diag}(J_k) F$ with F the unitary DFT and $J_k = \mathbb{E}|N_k|^2$ the two-sided bin power of the unitary-DFT coefficients $N_k = (Fn)_k$. Writing $X_k = (Fx)_k$ and $S_k = (Fs_0)_k$, the quadratic form is

$$(x - As_0)^T \Sigma^{-1} (x - As_0) = \sum_{k=0}^{d-1} \frac{|X_k - AS_k|^2}{J_k}, \qquad N_\Phi := s_0^T \Sigma^{-1} s_0 = \sum_{k=0}^{d-1} \frac{|S_k|^2}{J_k}, \tag{S1}$$

where both sums run over all d bins. On the one-sided rFFT grid the same quantities are obtained with endpoint weights, $\sum_{k=0}^{d-1}(\cdot) = \sum_{k=0}^{\lfloor d/2 \rfloor} w_k(\cdot)$ with $w_k = 1$ for the DC bin and (even d) the Nyquist bin and $w_k = 2$ for the interior bins, because $X_{d-k} = X_k^*$ and $J_{d-k} = J_k$; conversion of J_k to a physical one-sided PSD density is the factor recorded in Appendix S17. Every statement below is made in this convention. The proper-complex convention, in which each bin is an independent circular complex observation, is stated separately at the end, because a real trace's full DFT is *not* a vector of d independent proper complex bins.

Unbiasedness. The estimator is $\hat{A} = (s_0^T \Sigma^{-1} x)/N_\Phi = \sum_k \Phi_k^{(0)} X_k$ with $\Phi_k^{(0)} = S_k^*/(J_k N_\Phi)$, the two-sided sum being real by conjugate symmetry. Under $X_k = AS_k + N_k$,

$$\mathbb{E}[\hat{A}] = \sum_k \Phi_k^{(0)} A S_k = A \sum_k \frac{S_k^* S_k}{J_k N_\Phi} = A \cdot \frac{N_\Phi}{N_\Phi} = A. \tag{S2}$$

Estimator variance. In detector coordinates, $\hat{A} - A = s_0^T \Sigma^{-1} n / N_\Phi$, so

$$\operatorname{Var}(\hat{A}) = \frac{s_0^T \Sigma^{-1} \Sigma \Sigma^{-1} s_0}{N_\Phi^2} = \frac{1}{N_\Phi} = \Big(\sum_{k=0}^{d-1} \frac{|S_k|^2}{J_k} \Big)^{-1}, \tag{S3}$$

which is Eq. (8) of the main text. The same result follows bin by bin from $\operatorname{Var}(\hat{A}) = \sum_k |\Phi_k^{(0)}|^2 J_k$ provided the sum is the two-sided one: the conjugate pairs (N_k, N_{d-k}) are not independent, but their contributions to the real linear functional $\hat{A}$ add to exactly the two-sided formula.

Fisher information and the CRB. The log-likelihood of the real trace is $\log \mathcal{L} = -\frac{1}{2}(x - As_0)^T \Sigma^{-1}(x - As_0) + \text{const}$, so

$$\mathcal{I}(A) = -\mathbb{E}\left[\frac{\partial^2 \log \mathcal{L}}{\partial A^2}\right] = s_0^T \Sigma^{-1} s_0 = \sum_{k=0}^{d-1} \frac{|S_k|^2}{J_k} = N_\Phi, \tag{S4}$$

and $\mathrm{Var}(\hat{A}) = 1/\mathcal{I}(A)$: the CRB is attained in the same model in which the variance was computed, so the OF estimator is efficient and hence the minimum-variance unbiased estimator of the real-amplitude Gaussian model.

The proper-complex convention. If instead the observation is a vector of m independent proper (circular) complex bins, $X_k = AS_k + N_k$ with $\mathbb{E}|N_k|^2 = J_k$ and $\mathbb{E}[N_k^2] = 0$, and A is *real*, then the ML estimator is $\hat{A} = \mathrm{Re}\big(\sum_k S_k^* X_k / J_k\big) / \sum_k |S_k|^2/J_k$, with $\mathrm{Var}(\hat{A}) = 1/(2\sum_k |S_k|^2/J_k)$ and $\mathcal{I}(A) = 2\sum_k |S_k|^2/J_k$, again attaining the bound. If A is *complex*, the ML estimator is $\hat{A} = \sum_k S_k^* X_k/J_k \big/ \sum_k |S_k|^2/J_k$ with $\mathbb{E}|\hat{A} - A|^2 = 1/\sum_k |S_k|^2/J_k$, and the Fisher matrix for $(\mathrm{Re}\, A, \mathrm{Im}\, A)$ is $2\sum_k |S_k|^2/J_k \cdot I_2$. These are the statements one obtains by treating the positive-frequency bins of a real trace as independent, and they differ from (S3)–(S4) by exactly the factor of two that the conjugate-symmetric pairs contribute; the real convention above is the one that describes a real time trace.

Energy resolution. With amplitude calibrated to energy, $A = E/E_0$ for a known calibration energy E_0 (e.g. a K-α X-ray), the baseline energy resolution is

$$\sigma_E = E_0 \cdot \sigma_A = \frac{E_0}{\sqrt{N_\Phi}} = E_0 \left(\sum_k \frac{|S_k|^2}{J_k}\right)^{-1/2}, \tag{S5}$$

which is Eq. (8). Resolution improves when signal power is concentrated at frequencies where the noise PSD is small; N_Φ is both the filter normalization and the total signal-to-noise budget.

Bias under template mismatch. When the assumed template s_0 differs from the true signal s,

$$\mathbb{E}[\hat{A}_{\mathrm{OF}}] = A \cdot \frac{s_0^\dagger \Sigma^{-1} s}{s_0^\dagger \Sigma^{-1} s_0} = A \cdot \frac{\|\tilde{s}\|}{\|\tilde{s}_0\|}\, \rho, \qquad \rho = \frac{\tilde{s}_0^\dagger \tilde{s}}{\|\tilde{s}_0\|\, \|\tilde{s}\|}, \tag{S6}$$

with $|\rho| = \cos\theta_w$. Two effects are mixed in this factor and should be kept apart. A *normalization* mismatch, $s = \gamma s_0$ with $\gamma > 0$ real, gives $\rho = 1$ and $\mathbb{E}[\hat{A}] = \gamma A$: the amplitude calibration changes, with no loss of SNR and no bias toward zero in particular (for a signed or complex γ, $\rho = \gamma/|\gamma|$ carries the sign or phase, and ρ is a signed correlation, not the cosine of a principal angle). A *direction* mismatch, $\|\tilde{s}\| = \|\tilde{s}_0\|$ and $|\rho| < 1$, scales

the expectation by the signed (real case) or complex ρ: the attenuation magnitude is $|\rho| = \cos\theta_w$, and a negative real ρ reverses the sign of the response. In general the factor can be larger or smaller than one, so "biased toward zero" is not the right description. The variance formula (S3) is then no longer the CRB for the true signal; the effective SNR retention is

$$\frac{|s_0^\dagger \Sigma^{-1} s|^2}{(s_0^\dagger \Sigma^{-1} s_0)(s^\dagger \Sigma^{-1} s)} = \cos^2\theta_w, \tag{S7}$$

the squared cosine of the whitened angle between s_0 and s (modulus squared in the complex case). In whitened coordinates, with $u = \tilde{s}_0/\|\tilde{s}_0\|$ the unit template direction and $\hat{R}_{\tilde{x}} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, the OF projector retains the fraction $u^\dagger \hat{R}_{\tilde{x}} u / \operatorname{tr}\hat{R}_{\tilde{x}}$ of the *observed* whitened power (signal plus noise), a quantity invariant under rescaling of the template; the corresponding signal-only fraction is $u^\dagger R_t u / \operatorname{tr} R_t$ with R_t the whitened signal second moment of Proposition 2.4, and neither is a signal-to-noise ratio. When the true family is not one-dimensional, the remaining signal power is distributed across additional directions that rank-k EMPCA can retain.

Reference implementation. For completeness, the single-event OF amplitude estimator in the two-sided convention of (S1) is:

1. Compute *unitary* DFTs $X_k \leftarrow d^{-1/2}\,\mathrm{FFT}(x)$, $S_k \leftarrow d^{-1/2}\,\mathrm{FFT}(s_0)$, $k = 0, \ldots, d-1$, so that $J_k = \mathbb{E}|N_k|^2$ is the bin power of the same unitary transform (with an unnormalized FFT, J_k must be replaced by dJ_k).

2. Numerator $\mathcal{N} \leftarrow \mathrm{Re}\big(\sum_{k=0}^{d-1} S_k^* X_k / J_k\big)$ (the real part removes rounding residue only; the two-sided sum is real).

3. Denominator $\mathcal{D} \leftarrow \sum_{k=0}^{d-1} |S_k|^2/J_k = N_\Phi$.

4. $\hat{A} \leftarrow \mathcal{N}/\mathcal{D}$; residual $\chi^2 \leftarrow \sum_{k=0}^{d-1} |X_k - \hat{A}S_k|^2/J_k$; predicted variance $1/N_\Phi$.

All operations are $O(d)$ after the one-time $O(d \log d)$ FFT. The amplitude ratio $\mathcal{N}/\mathcal{D}$ is invariant under a common rescaling of J_k and therefore cannot reveal a normalization mismatch; χ^2, N_Φ and the predicted variance are not, and they must equal their time-domain values $(x - \hat{A}s_0)^T\Sigma^{-1}(x - \hat{A}s_0)$, $s_0^T\Sigma^{-1}s_0$ and $1/(s_0^T\Sigma^{-1}s_0)$ under one stated convention (the regression check `scripts/check_revision_claims.py` compares all four quantities between the time-domain, two-sided and endpoint-weighted rFFT forms). An implementation on the rFFT grid replaces each sum by its endpoint-weighted one-sided form, and an implementation that stores J_k as a physical one-sided PSD density carries the conversion factor of Appendix S17 in numerator, denominator and residual alike. The normalization actually implemented in the experiment repository is checked against the

bound (S3) by the controlled test S1 of Section 5. For time-shift OF, step 1 is augmented by a grid search over candidate shifts with $S_k \to S_k e^{-2\pi i k \tau_j / N}$, selecting the shift that minimizes χ^2 (Appendix S7.1).

S6 Shift and joint-channel optimal filters

S7 Time-Shift and Joint-Channel Optimal Filtering

This appendix gives the full derivations for the two OF extensions summarized in Section 4.1, and proves the degeneracy propositions referenced there.

S7.1 Time-shift optimal filtering

Include an unknown arrival-time shift t_0 via

$$S_k(t_0) = e^{-i\omega_k t_0} S_k, \qquad X_k = A\, S_k(t_0) + N_k, \tag{S8}$$

with weighted objective

$$\chi^2(A, t_0) = \sum_k \frac{|X_k - A e^{-i\omega_k t_0} S_k|^2}{J_k}. \tag{S9}$$

For fixed t_0, the stationarity condition $\partial\chi^2/\partial A^* = 0$ yields

$$\hat{A}(t_0) = \frac{\sum_k e^{i\omega_k t_0} S_k^* X_k / J_k}{\sum_k |S_k|^2/J_k} = \sum_k e^{i\omega_k t_0} \Phi_k^{(0)} X_k, \tag{S10}$$

so $\hat{A}(t_0)$ is the inverse DFT of the filtered spectrum [7]. To profile out the amplitude, define

$$\chi_0^2 = \sum_k \frac{|X_k|^2}{J_k}, \qquad B(t_0) = \sum_k \frac{e^{i\omega_k t_0} S_k^* X_k}{J_k}. \tag{S11}$$

Then

$$\chi^2(A, t_0) = \chi_0^2 - A^* B(t_0) - A B^*(t_0) + |A|^2 N_\Phi, \tag{S12}$$

and at $A = \hat{A}(t_0) = B(t_0)/N_\Phi$,

$$\chi^2(t_0) = \chi_0^2 - \frac{|B(t_0)|^2}{N_\Phi} = \chi_0^2 - |\hat{A}(t_0)|^2 N_\Phi. \tag{S13}$$

Hence the optimal delay is

$$\hat{t}_0 = \arg\min_{t_0} \chi^2(t_0) = \arg\max_{t_0} |\hat{A}(t_0)|, \tag{S14}$$

the usual "maximize filtered amplitude in time" criterion for real-valued traces with Hermitian spectra.

Relation to learned subspaces. The shifted templates $\mathcal{M} = \{e^{-i\omega t_0} s_0 : t_0 \in \mathbb{R}\}$ form a one-parameter signal manifold, and time-shift OF performs ML estimation over this manifold. For small shifts, the Taylor expansion

$$e^{-i\omega t_0} s_0 = s_0 - i\omega t_0\, s_0 - \tfrac{1}{2}(\omega t_0)^2 s_0 + \cdots \tag{S15}$$

shows that the shift family is approximately contained in $\mathrm{span}\{s_0, \omega s_0, \omega^2 s_0, \dots\}$ — in time-domain notation, $\mathrm{span}\{s_0, \partial_t s_0, \partial_t^2 s_0, \dots\}$.

Remark S7.1 (Shifted OF and learned subspaces)**.** *OF with time shift is the ML estimator obtained by restricting the signal model to the nonlinear one-parameter manifold $\mathcal{M}$. Its relation to a learned linear subspace is one of approximation, not degeneracy: the linear span of $\mathcal{M}$ is generically all of $\mathbb{C}^d$, so there is no rank at which a subspace* equals *the shift family, and a rank-k EMPCA subspace can only contain the truncated Taylor family* (S15) *to some order. When the dominant variation in the data is timing jitter that is small on the scale of the template's bandwidth, the leading learned directions are expected to approximate* $\mathrm{span}\{s_0, \partial_t s_0, \dots\}$ *and rank-k EMPCA then acts as a linear-subspace approximation to shifted OF; this is a heuristic expectation about the data, and no precise hypothesis under which it holds is claimed here.*

Basic OF, shifted OF, and rank-k EMPCA are thus related in the informal sense that each enlarges the admissible family of the previous one, with the last relation an approximation rather than a containment. The former numerical illustration has been omitted from this revision because its seed-level provenance was not recoverable from the archived evidence; the analytical statement above does not depend on it.

S7.2 Joint-channel correlated optimal filtering

Joint observation model. For C channels with d samples per channel, stack the channel traces:

$$x = \begin{bmatrix} x^{(1)} \\ \vdots \\ x^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \qquad s = \begin{bmatrix} s^{(1)} \\ \vdots \\ s^{(C)} \end{bmatrix} \in \mathbb{C}^{Cd}, \tag{S16}$$

with noise $n \sim \mathcal{N}_{\mathbb{C}}(0, \Sigma_{\text{joint}})$ and block-structured covariance

$$\Sigma_{\text{joint}} = \begin{bmatrix} \Sigma^{(11)} & \Sigma^{(12)} & \dots & \Sigma^{(1C)} \\ \Sigma^{(21)} & \Sigma^{(22)} & \dots & \Sigma^{(2C)} \\ \vdots & & \ddots & \vdots \\ \Sigma^{(C1)} & \dots & \dots & \Sigma^{(CC)} \end{bmatrix}, \tag{S17}$$

where $\Sigma^{(cc)}$ is the intra-channel covariance and $\Sigma^{(cc')}$ the cross-channel correlation; for stationary noise, $\Sigma^{(cc)}$ is circulant and $\Sigma^{(cc')}$ is the cross-spectral density matrix.

Estimator and degeneracy. The Gaussian ML objective is unchanged in form, $\chi^2(\hat{s}) = (x - \hat{s})^\dagger \Sigma_{\text{joint}}^{-1}(x - \hat{s})$. The rank-1 constraint $\hat{s} = As$ gives the joint-channel correlated optimal filter

$$\hat{A}_{\text{JOF}} = \frac{s^\dagger \Sigma_{\text{joint}}^{-1} x}{s^\dagger \Sigma_{\text{joint}}^{-1} s}, \tag{S18}$$

which is algebraically the rank-1 weighted subspace estimator (9) with $U = s$.

Proposition S7.1 (Degeneracy to joint-channel correlated OF)**.** *Weighted EMPCA with the full inverse covariance $\Sigma_{\text{joint}}^{-1}$ is the multi-channel subspace generalization of joint-channel correlated optimal filtering. In the rank-1 case, when the EMPCA basis spans the joint template direction, weighted EMPCA reduces exactly to joint-channel correlated OF.*

EMPCA does not change the underlying likelihood; it replaces a fixed joint template by a learned joint signal subspace, with noise correlations handled identically through the quadratic form.

Whitening and decoupling. The joint whitening map is any W with $W\Sigma_{\text{joint}}W^\dagger = I$, e.g. the Hermitian $\Sigma_{\text{joint}}^{-1/2}$ from an eigendecomposition or L^{-1} from a Cholesky factor (Appendix S9.8). For stationary noise with per-frequency cross-spectral matrices $\Sigma_f \in \mathbb{C}^{C\times C}$, whitening decomposes into independent $C \times C$ factors per frequency bin, reducing to C scalar divisions when channels are uncorrelated. If $\Sigma^{(cc')} = 0$ for $c \neq c'$, the joint objective is the sum of the C per-channel quadratic forms. What this implies depends on the signal model. For joint-channel OF with a fixed joint template the amplitude is shared, and the estimator is the Fisher-information-weighted mean of the per-channel OF estimates. For a learned rank-k subspace of $\mathbb{C}^{Cd}$ the objective does *not* decouple into independent per-channel subspace problems: the optimal joint subspace is the top-k eigenspace of the whitened joint second moment, whose cross-channel blocks are set by the signal, and a latent that excites two channels couples them however independent the noise is (with $\Sigma_{\text{joint}} = I$ and a signal second moment $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ across two single-sample channels, the rank-one joint direction is $(1, 1)/\sqrt{2}$ and uses both). Independent single-channel EMPCA is the further restriction of the class to block-diagonal bases with per-channel coefficients, which is an assumption about the signal, not a consequence of independent noise.

S8 EMPCA implementation, convergence and conditioning

S9 EMPCA Updates, Convergence, and Conditioning

This appendix collects the algorithmic content supporting Section 4.2: the EM updates and their derivation, the frequency-domain variant, the coordinate-descent identity with the AE objective, the convergence proof, and the practical machinery (initialization, model-order selection, whitening numerics, and conditioning).

S9.1 Classical PCA and its noise assumption

For a mean-subtracted data matrix, classical PCA seeks a rank-k model $X \approx \Gamma^\dagger U^\dagger$ (rows of X are events) minimizing the Euclidean reconstruction error $\|X - \Gamma^\dagger U^\dagger\|_F^2$ subject to $U^\dagger U = I_k$; the solution is the span of the top-k eigenvectors of the sample covariance. PCA is the ML subspace estimator only under isotropic noise $\Sigma = \sigma^2 I$; any colored or heteroskedastic noise breaks this optimality. EMPCA [13] restores ML optimality by carrying the weight Σ^{-1} (or per-observation weights) through the estimation.

S9.2 EM updates

Write the joint objective in the variables (U, Γ), with $\Gamma = [c_1, \ldots, c_N]$ the coefficient matrix,

$$\mathcal{L}(U, \Gamma) = \frac{1}{N} \sum_{i=1}^{N} (x_i - U c_i)^\dagger \Sigma^{-1} (x_i - U c_i), \tag{S19}$$

which is also the tied-AE loss of Section 4.3 with the encoder constraint imposed at convergence.

E-step (coefficients). For fixed U, $\mathcal{L}$ is strictly convex and quadratic in each c_i; the unique minimizer solves the weighted normal equations

$$(U^\dagger \Sigma^{-1} U)\, c_i = U^\dagger \Sigma^{-1} x_i, \qquad \hat{c}_i = (U^\dagger \Sigma^{-1} U)^{-1} U^\dagger \Sigma^{-1} x_i, \tag{S20}$$

which reduces to $\hat{c}_i = U^\dagger \Sigma^{-1} x_i$ under the Σ^{-1}-orthonormality constraint — exactly the tied-AE encoder.

M-step (basis). For fixed Γ, with aggregates $G = \Gamma\Gamma^\dagger/N$ and $H = \Gamma X/N$ (so that $H^\dagger = N^{-1}\sum_i x_i c_i^\dagger$), the unconstrained minimizer (unique when $G \succ 0$) is

$$U_{\text{unc}} = H^\dagger G^{-1} = \Big(\frac{1}{N}\sum_i x_i c_i^\dagger\Big)\Big(\frac{1}{N}\sum_i c_i c_i^\dagger\Big)^{-1}, \tag{S21}$$

followed by re-orthonormalization in the Σ^{-1} inner product *with the compensating change of coefficients*: with the thin QR factorization $\Sigma^{-1/2}U_{\text{unc}} = QR$, set

$$U_{\text{new}} = \Sigma^{1/2}Q, \qquad \Gamma_{\text{new}} = R\,\Gamma, \qquad \text{so that} \qquad U_{\text{new}}\Gamma_{\text{new}} = U_{\text{unc}}\Gamma. \tag{S22}$$

The normalization is then a change of gauge that leaves the product, and hence the objective, unchanged; the next E-step recomputes the coefficients from U_{new} in any case. Normalizing U with Γ held fixed is *not* a descent step: for the scalar example $x = 2$, $c = 1$, $U_{\text{unc}} = 2$ the loss is zero before normalization and one after replacing U by 1 with c fixed. The weight Σ^{-1} does not appear in (S21): the stationarity condition of the weighted objective in U is $\Sigma^{-1}\sum_i (x_i - Uc_i)c_i^\dagger = 0$, and Σ^{-1} is invertible (an update that carries an extra Σ^{-1} in H is monotone but converges to a different subspace). Substituting the E-step coefficients $c_i = U^\dagger\Sigma^{-1}x_i$ gives $U_{\text{unc}} = \hat{R}_x\Sigma^{-1}U\,G^{-1}$, so in whitened coordinates $W = \Sigma^{-1/2}U$ the update is $\text{span}(W_{\text{new}}) = \text{span}(\hat{R}_{\tilde{x}}W)$ with $\hat{R}_{\tilde{x}} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$: EMPCA with exact block steps is orthogonal (subspace) iteration on the whitened second moment [17], which is what its convergence theory below rests on.

Coordinate-descent view. Each step is the exact block minimizer of (S19) over its block, the M-step being taken over unconstrained U and followed by the gauge change (S22), so

$$\mathcal{L}(U^{(r+1)}, \Gamma^{(r+1)}) \le \mathcal{L}(U^{(r)}, \Gamma^{(r+1)}) \le \mathcal{L}(U^{(r)}, \Gamma^{(r)}), \tag{S23}$$

where $(U^{(r+1)}, \Gamma^{(r+1)})$ denotes the normalized pair $(U_{\text{new}}, \Gamma_{\text{new}})$ of (S22), whose objective equals that of $(U_{\text{unc}}, \Gamma^{(r+1)})$; i.e. the EMPCA E- and M-steps are block-coordinate descent on the noise-aware AE loss. The objective is convex in each block separately, but the feasible set of the constrained basis (an orthonormality constraint) is not convex, so the problem is not biconvex in the usual sense; the descent argument uses only the exact block minimizations. This is the algorithmic face of the Bridge Theorem. Equivalently one may drop the coefficients after the M-step and describe the algorithm as the *profiled* subspace iteration $W \mapsto \text{orth}(\hat{R}_{\tilde{x}}W)$ on the whitened second moment, which is what the convergence theory below analyses.

Three algorithms, one optimum. Three procedures compute or approximate the minimizer of (S19), and they are not interchangeable. (a) A direct eigendecomposition

of $\hat{R}_{\tilde{x}}$, or a thin SVD of the whitened data matrix, returns the top-k eigenspace in one step; this is how the archived weighted-PCA fits of Section 5 are computed, and it needs no convergence theory. (b) The common-covariance EMPCA iteration above with exact block steps is orthogonal iteration on $\hat{R}_{\tilde{x}}$, and the convergence statements of this appendix are statements about it. (c) Heterogeneous-weight EMPCA [13], with per-event or per-element weights Σ_i^{-1} or ω_{kj} for which no single whitening exists, is a genuinely alternating scheme: it retains the monotone descent (S23) but is not a subspace iteration on a fixed matrix, and the identification with (a)–(b) does not apply to it.

Deflation variant. Equivalent to Bailey's one-component deflation, components may be solved sequentially, updating $X \leftarrow X - u_m c_m$ before solving the next component.

Reference iteration. Rank-k EMPCA with weighting Σ^{-1}:

1. Initialize $U^{(0)}$ (random orthonormal, SVD warm-start, or template-seeded; see below).
2. E-step: solve (S20) for all events ($O(k^2 dN)$).
3. M-step: form (S21); re-orthonormalize via QR of $\Sigma^{-1/2}U_{\text{unc}}$ and transport the coefficients, $\Gamma \leftarrow R\Gamma$, as in (S22) ($O(kdN) + O(k^2 d)$); or discard Γ and let the next E-step recompute it (profiled iteration).
4. Evaluate $\mathcal{L}^{(r+1)}$; stop when the relative decrease is below tolerance.

For $k = 1$ the iteration is the power method on $\hat{R}_{\tilde{x}}$. From a template seed it starts near the fixed point whenever the leading sample eigenvector is close to the whitened template, but the seed is not preserved: one step already rotates it toward the sample eigenvector (for $\hat{R}_{\tilde{x}} = \text{diag}(2, 1)$ the seed $(1, 1)/\sqrt{2}$ moves to $(2, 1)/\sqrt{5}$, by $18.4°$), so the number of iterations to a given tolerance depends on the sample spectrum and no finite-sample guarantee of convergence in $O(1)$ iterations follows from the seed.

S9.3 Frequency-domain variant and dual interpretation

Yu's microcalorimeter setting [14] motivates a frequency-domain variant robust to random time shifts. In Fourier space, $X_k^{(j)} = A_j e^{-i\omega_k t_{0,j}} S_k + N_k^{(j)}$, and direct PCA is confounded because the effective coefficient depends on the frequency bin. Using phase differences $\Delta\phi_k = \phi_k - \phi_{k-1}$, a time shift contributes $(\omega_k - \omega_{k-1})t_0$, which is constant for uniform frequency spacing; the transformed coefficients then behave as a single complex factor per event, restoring a linear subspace model.

With per-element weights ω_{kj} (e.g. inverse PSD; the symbol W is reserved for the whitened basis), the weighted EMPCA objective is

$$\chi^2_{\mathrm{EMPCA}} = \sum_k \sum_j \omega_{kj} \left| X_{kj} - [U\Gamma]_{kj} \right|^2 , \tag{S24}$$

with weighted E-step normal equations $(U^\dagger \Omega_j U) c_j = U^\dagger \Omega_j X_{:,j}$ per event, where $\Omega_j = \mathrm{diag}_k(\omega_{kj})$, and weighted M-step row solves $(\Gamma \Omega_k \Gamma^\dagger) u_k = \Gamma \Omega_k X^\dagger_{k,:}$ per frequency, where $\Omega_k = \mathrm{diag}_j(\omega_{kj})$. For stationary noise the covariance is diagonal in the Fourier basis, Σ^{-1} acts element-wise, and weighted inner products reduce to Hadamard products: the Hadamard-product implementation is the exact dual PCA formulation under a diagonal noise covariance, not an approximation.

S9.4 Amplitude estimation details

Gauge-invariant amplitude estimators. Because the basis is identified only up to $\mathcal{U}(k)$ rotation, physical amplitudes use gauge-invariant functionals: the whitened reconstruction norm $\|\hat{s}\|_{\Sigma^{-1}} = \|c\|_2$ (under $U^\dagger \Sigma^{-1} U = I_k$), calibrated against a reference line; or the projection onto a calibration template, Eq. (4.2).

Ridge-regularized coefficient solves (fixed basis). Two Gram matrices must be kept apart. The *basis* Gram matrix $U^\dagger \Sigma^{-1} U$ is ill-conditioned when basis vectors are nearly collinear in whitened space; it is I_k for a Σ^{-1}-orthonormal basis whatever the rank of the data. The *coefficient* Gram matrix $\Gamma\Gamma^\dagger/N$ has rank at most $\mathrm{rank}\, X$, so it is singular whenever k exceeds the data rank, and no coefficient regularization changes that. For a *fixed* basis with a nearly collinear basis Gram matrix, Tikhonov regularization

$$\hat{c}_\mu = (U^\dagger \Sigma^{-1} U + \mu I_k)^{-1} U^\dagger \Sigma^{-1} x \tag{S25}$$

controls the bias–variance tradeoff; the ridge parameter μ (distinct from the eigenvalues λ_j) may be chosen by generalized cross-validation [50], though for $N \gg k$ a small numerical floor suffices after proper re-orthonormalization.

S9.5 Convergence proof

Theorem S9.1 (Monotone convergence of EMPCA)**.** *Consider the common-covariance iteration of Appendix S9.2 with exact E- and M-steps and the gauge transport* (S22)*, and write* $W^{(r)} = \Sigma^{-1/2} U^{(r)}$ *for the whitened basis. Assume* $\mathrm{rank}\, \hat{R}_{\tilde{x}} \geq k$ *and that the initial basis satisfies* $\mathrm{rank}(\hat{R}_{\tilde{x}} W^{(0)}) = k$*, i.e.* $\mathrm{range}\, W^{(0)} \cap \ker \hat{R}_{\tilde{x}} = \{0\}$*, which holds for almost every* $W^{(0)}$ *(Haar measure on the Stiefel manifold). Then the coefficient Gram matrix*

$G^{(r)} = \Gamma^{(r)}\Gamma^{(r)\dagger}/N$ is positive definite at every iteration, so the M-step (S21) *is defined and unique throughout, and: (i) $\mathcal{L}^{(r+1)} \le \mathcal{L}^{(r)}$ for all r; (ii) the sequence $\{(U^{(r)}, \Gamma^{(r)})\}$ has at least one limit point, and G is positive definite at every limit point; (iii) every limit point is a fixed point of the alternating-minimization map, and hence satisfies the first-order stationarity conditions of* (9) *in the projector variable $UU^\dagger\Sigma^{-1}$. When $\operatorname{rank}\hat{R}_{\tilde{x}} < k$ the coefficient Gram matrix is singular from the first iteration, the M-step is undefined, and the iteration as written must stop with a recorded rank failure or be restarted with $k \le \operatorname{rank}\hat{R}_{\tilde{x}}$; a ridge on the coefficient solve does not remove the singularity (Appendix S9.4), and no regularized alternating algorithm is defined or analysed here.*

Proof. With the E-step coefficients $\Gamma = W^\dagger\tilde{X}^\dagger$ the Gram matrix is $G = W^\dagger\hat{R}_{\tilde{x}}W$, which is positive definite exactly when $\operatorname{range} W \cap \ker\hat{R}_{\tilde{x}} = \{0\}$; this is the assumption at $r = 0$. For $r \ge 1$, $\operatorname{range} W^{(r)} = \hat{R}_{\tilde{x}} \operatorname{range} W^{(r-1)} \subseteq \operatorname{range}\hat{R}_{\tilde{x}}$ (Appendix S9.2), on which $\hat{R}_{\tilde{x}}$ is bounded below by its smallest nonzero eigenvalue $\lambda^+_{\min}$, so $G^{(r)} \succeq \lambda^+_{\min} I_k$ for all $r \ge 1$; the set of bases with range inside $\operatorname{range}\hat{R}_{\tilde{x}}$ is closed, so the same bound holds at every limit point. This is the Gram condition used below, and it is not an extra hypothesis. The E-step solution (S20) is the unique global minimizer of a strictly convex quadratic, so $\mathcal{L}(U^{(r)}, \Gamma^{(r+1)}) \le \mathcal{L}(U^{(r)}, \Gamma^{(r)})$. The M-step minimizes the quadratic over unconstrained U, and the re-orthonormalization with coefficient transport (S22) leaves the product $U\Gamma$, hence the objective, unchanged. Chaining gives monotone decrease (S23); since $\mathcal{L} \ge 0$, the sequence converges. The iterates live in a compact set (orthonormality constraint plus bounded coefficients), so a limit point exists. For (iii): on the set where $G \succ 0$ the M-step (S21) is a continuous function of Γ, the thin QR re-orthonormalization is continuous up to the $\mathcal{U}(k)$ gauge, and the E-step is continuous in U, so the update map $F : (U, \Gamma) \mapsto (U^{(r+1)}, \Gamma^{(r+1)})$ is continuous there up to gauge. Along a subsequence converging to a limit point (U^*, Γ^*) with $G^* \succ 0$, $\mathcal{L}(F(U^*, \Gamma^*)) = \lim \mathcal{L}^{(r+1)} = \mathcal{L}(U^*, \Gamma^*)$; because the E-step minimizer is unique, equality in the E-step forces Γ^* to be the E-step of U^*, and because the M-step minimizer is unique when $G^* \succ 0$, equality in the M-step forces U^* to be the M-step of Γ^* up to gauge. Thus $F(U^*, \Gamma^*) = (U^*, \Gamma^*)$ up to gauge, and a fixed point of both exact block minimizations satisfies the first-order conditions of (9). The initial condition is a genericity condition on $W^{(0)}$: it fails on the null set of initial subspaces that meet $\ker\hat{R}_{\tilde{x}}$, and it cannot hold at all when k exceeds the data rank. □

Remark S9.1 (Stationary points and almost-everywhere convergence)**.** *The objective is non-convex jointly in (U, Γ) for $k \ge 2$, and its stationary points are* not *all equivalent. In the projector variable the problem is the Rayleigh–Ritz maximization of $\operatorname{Tr}(W^\dagger\hat{R}_{\tilde{x}}W)$ over $W^\dagger W = I_k$ (Appendix S11.3), whose stationary points are all k-dimensional invariant subspaces of $\hat{R}_{\tilde{x}}$ — any choice of k eigenvectors when the eigenvalues are distinct — and only the top-k one is the global minimizer of $\mathcal{L}$. Rotating a retained eigenvector v_i*

toward an unretained v_j, $w(\theta) = \cos\theta\, v_i + \sin\theta\, v_j$, changes the loss by $\mathcal{L}(\theta) - \mathcal{L}(0) = (\lambda_i - \lambda_j)\sin^2\theta$. Hence, for distinct eigenvalues and $0 < k < d$, an invariant subspace is a local minimum of $\mathcal{L}$ if and only if every retained eigenvalue exceeds every unretained one (the top-k subspace), a local maximum *if and only if every retained eigenvalue is below every unretained one (the bottom-k subspace; for $k = 1$ and $\hat{R}_{\tilde{x}} = \mathrm{diag}(2, 1)$ the loss along $w = (\sin\theta, \cos\theta)$ is $2 - \sin^2\theta$, maximal at the bottom eigenvector), and a saddle otherwise [30]. With repeated eigenvalues the critical sets are manifolds and the classification is by the same inequalities applied to the multiset; $k = 0$ and $k = d$ are trivial. This classification does not by itself say where the alternating algorithm goes. What does is the identification of the exact-block-step iteration with orthogonal iteration on $\hat{R}_{\tilde{x}}$ (Appendix S9.2): when $\lambda_k > \lambda_{k+1}$, the iterates converge to the top-k invariant subspace for every initial W whose projection onto that subspace is nonsingular — almost every W — at the linear rate λ_{k+1}/λ_k [17, Thm. 8.2.2], and the coefficient-Gram condition of Theorem S9.1 holds along the way. The bottom-k and mixed invariant subspaces are reached only from the null set of initial subspaces orthogonal to some top eigenvector. Multiple random restarts (Appendix S9.6) remain the check used in this paper, and the guarantee does* not *transfer to the alternating scheme of the separable class (Section 4.4), which is not a subspace iteration on a fixed matrix.*

Remark S9.2 (Rate)**.** *When $\lambda_{k+1} < \lambda_k$, the iteration converges at the linear rate λ_{k+1}/λ_k of orthogonal iteration [17], consistent with the local rate that block-coordinate-descent theory gives for the quadratic objective [16].*

S9.6 Initialization

Random: draw $U^{(0)}$ Haar-uniformly on the Stiefel manifold; simple, effective when $N \gg kd$; multiple restarts check convergence to the same subspace (Remark S9.1). *SVD warm-start:* top-k right singular vectors of the unweighted data matrix; a good warm start when noise is not far from isotropic. *Template-seeded:* set $u_1 \propto \tilde{s}/\|\tilde{s}\|$ from a calibration template, extending to rank k with random whitened-orthogonal directions. This starts the iteration inside the containment condition of Section 4.2 but does not preserve it: the iteration moves the seed toward the sample eigenvectors (Appendix S9.2), so containment after learning holds only when it is imposed as a constraint — fixing u_1 and learning the remaining $k - 1$ directions in its whitened orthogonal complement — and is otherwise an empirical property of the fit.

S9.7 Model-order selection

The rank is a modelling choice with a stated target, and the target has to be one that a held-out residual can estimate. The held-out self-reconstruction residual is *not* such a criterion: for nested fitted subspaces the whitened residual of every trace, held out or not, is nonincreasing in k and reaches zero at $k = d$, so "the k at which the out-of-sample χ^2 stops decreasing" rewards retaining noisy dimensions and is not a denoising-rank rule. Three criteria with stated assumptions are used instead.

Corrected held-out risk. For a projector fitted on training data and evaluated on held-out traces with a correct Σ, Proposition 2.1 gives the whitened clean-signal risk as $\mathcal{R}_{\text{clean},\Sigma}(k) = \mathcal{R}_{\text{obs}}(k) + 2k - d$, so the rank that minimizes the population clean-signal risk within the projector class is estimated by

$$\hat{k} = \arg\min_k \big[\hat{\mathcal{R}}_{\text{obs}}(k) + 2k\big], \tag{S26}$$

the held-out mean whitened residual per trace plus the degrees-of-freedom penalty $2k$. Its population interpretation is transparent in the ideal nested case: if the fitted subspaces are the top-k eigenspaces of the whitened signal second moment R_t with eigenvalues $\mu_1 \geq \mu_2 \geq \cdots$, then $\mathcal{R}_{\text{clean},\Sigma}(k) = \sum_{j>k} \mu_j + k$ and the increment from k to $k+1$ is $1 - \mu_{k+1}$, so the population optimum retains exactly the signal eigenvalues above one (ties at $\mu_{k+1} = 1$ are indifferent). Pure noise ($\mu \equiv 0$) has population optimum $k = 0$, which a finite validation sample estimates but does not guarantee; and "total signal energy above d" is only the condition for the identity map to beat the zero map, not a characterization of the optimal rank (with $\mu = (100, 0)$ and $d = 2$ the risks for $k = 0, 1, 2$ are $100, 1, 2$). The rule is a risk *estimate*: it is valid for fixed projectors evaluated on data they were not fitted to, requires the whitened noise second moment to be I (a wrong Σ biases the penalty), and compares projectors only; the zero and mean-only reconstructions ($k = 0$ with and without a fitted mean) are its baselines, and a shrinkage estimator uses the trace correction of Proposition 2.1. Rank selection and covariance accuracy are therefore coupled (Remark 3.4).

Noise-aware spectral gap. In whitened coordinates the noise floor has expected eigenvalue 1, and for N traces in d dimensions the sample noise eigenvalues spread up to the Marchenko–Pastur edge $(1 + \sqrt{d/N})^2$ (Section 4.2); a rank chosen as the number of whitened sample eigenvalues above that edge is a criterion with an explicit assumption (isotropic whitened noise, one-spike asymptotics used as a scale), which should be stated when it is used.

Probabilistic marginal likelihood. The model with event-specific fitted coefficients has a parameter count that grows with N, so the standard AIC/BIC penalties do not apply to it without a marginal likelihood that integrates the coefficients out. If that route is

taken, the subspace itself contributes $2k(d-k)$ real parameters (the dimension of the complex Grassmannian $\mathrm{Gr}(k,d)$, or $k(d-k)$ for real data) — not $2kd-k^2$, which is the dimension of the Stiefel manifold and still contains the k^2 gauge parameters — and the Laplace-approximated evidence of probabilistic PCA [51] is the reference criterion in isotropic whitened noise.

A physical upper bound on k (the number of independent pulse-shape variation modes) usefully narrows the search. Whichever criterion is used, the selection is made on development data and the selected rank is then evaluated once on a separate untouched split; the archived experiments of Section 5 fix the rank in the frozen configuration rather than selecting it.

S9.8 Whitening numerics and conditioning

Computing $\Sigma^{-1/2}$. For circulant Σ (stationary noise), whitening is element-wise division by $\sqrt{J_k}$ in the frequency domain. Otherwise: eigendecomposition $\Sigma = V\Lambda V^\dagger$ gives $\Sigma^{-1/2} = V\Lambda^{-1/2}V^\dagger$ (numerically stable, $O(d^3)$); for column observations and the Cholesky factorization $\Sigma = LL^\dagger$ the whitening map is $W = L^{-1}$, which satisfies $W\Sigma W^\dagger = I$ (the transposed choice $L^{-\dagger}$ does not: for $\Sigma = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ it gives $L^{-\dagger}\Sigma L^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ -1 & 2 \end{smallmatrix}\right)$). W is triangular, not Hermitian, and differs from $\Sigma^{-1/2}$ by a unitary factor, $W = Q\Sigma^{-1/2}$ with $Q = L^{-1}\Sigma^{1/2}$; every Σ^{-1} inner product, projector, residual and principal angle is the same under either map, so the choice is numerical (Cholesky is faster and preferred for well-conditioned dense Σ), but a nonsymmetric whitening map should be written W rather than $\Sigma^{-1/2}$; the archived fits use the eigendecomposition route.

Ill-conditioned noise PSDs. Bins with $J_k \approx 0$ amplify noise after whitening. Remedies: (i) an additive floor $J_k \to \max(J_k, \delta \max_k J_k)$ with $\delta \in [10^{-4}, 10^{-2}]$; (ii) frequency masking to a band $[f_{\min}, f_{\max}]$ where the signal has appreciable power — lossless when the signal has negligible power outside the band, and standard in cryogenic detector OF; (iii) Tikhonov regularization $\Sigma^{-1} \to (\Sigma + \varepsilon I)^{-1}$ when the noise floor itself is uncertain.

Rank deficiency and ill-conditioned bases. If the coefficient Gram matrix $\Gamma\Gamma^\dagger/N$ is singular (data rank below k, or an initialization meeting $\ker \hat{R}_{\tilde{x}}$), the iteration of Appendix S9.2 stops with a recorded rank failure and is restarted with a smaller k; this is the policy assumed by Theorem S9.1. A large condition number of the basis Gram matrix $U^\dagger\Sigma^{-1}U$ between re-orthonormalizations is a separate numerical symptom of near-collinear basis vectors, cured by re-orthonormalizing every M-step (standard); the coefficient ridge (S25) is a fixed-basis shrinkage device and is not part of the alternating algorithm.

Computational notes. Sliding-window inference uses fixed eigenvectors with fast coefficient solves, producing amplitude, χ^2, and likelihood traces efficiently. The E-step at $O(k^2dN)$ dominates at large k but is negligible relative to FFT cost at the small ranks ($k \leq 6$) used in this paper; batched GPU linear algebra (e.g. batched GEMM and fused weighting/projection kernels) accelerates batch workloads.

S10 Proofs

S11 Proofs of the Theorem Statements

S11.1 Extended proof of Theorem 4.2

We restate and prove the rank-1 OF–EMPCA equivalence in full detail. We work in whitened coordinates $\tilde{x}_i = \Sigma^{-1/2} x_i$, $\tilde{s} = \Sigma^{-1/2} s$, $\tilde{n}_i = \Sigma^{-1/2} n_i$, so that

$$\tilde{x}_i = a_i \tilde{s} + \tilde{n}_i, \qquad \tilde{n}_i \sim \mathcal{N}(0, I). \tag{S27}$$

Step 1: population second moment is a spiked identity. By assumption (ii) of the theorem, $\mathbb{E}[\tilde{n}_i \tilde{n}_i^\dagger] = I$ and the signal–noise cross moment vanishes, $\mathbb{E}[a_i \tilde{n}_i^\dagger] = 0$; by assumption (i), $s \neq 0$ and $\nu := \mathbb{E}|a_i|^2 > 0$. Then

$$R := \mathbb{E}[\tilde{x}_i \tilde{x}_i^\dagger] = \nu\, \tilde{s}\tilde{s}^\dagger + I. \tag{S28}$$

The eigenvalues of R are $\lambda_1 = 1 + \nu\|\tilde{s}\|^2$ (simple, eigenvector $\tilde{s}/\|\tilde{s}\|$) and $\lambda_j = 1$ for $j \geq 2$. All three assumptions are used: without $s \neq 0$ or $\nu > 0$ there is no spike, and a nonzero cross moment adds a term $\mathbb{E}[a_i \tilde{n}_i^\dagger]\tilde{s}^\dagger$ plus its adjoint that moves the leading eigenvector off $\tilde{s}$.

Step 2: EMPCA recovers the leading eigenvector. Rank-1 EMPCA maximizes the explained population second moment (the data are not centered; R is the uncentered second moment), i.e. finds the leading eigenvector of R:

$$u_1 = \frac{\tilde{s}}{\|\tilde{s}\|}. \tag{S29}$$

The eigenvalue gap $\lambda_1 - \lambda_2 = \nu\|\tilde{s}\|^2 > 0$ guarantees uniqueness whenever $\nu > 0$ and $s \neq 0$.

Step 3: the EMPCA projection coefficient. The raw projection coefficient is $u_1^\dagger \tilde{x}_i = \tilde{s}^\dagger \tilde{x}_i / \|\tilde{s}\|$. Rescaling so that a unit-amplitude signal gives $\hat{a}_i = 1$,

$$\hat{a}_i^{(\mathrm{EMPCA})} = \frac{\tilde{s}^\dagger \tilde{x}_i}{\tilde{s}^\dagger \tilde{s}} = \frac{s^\dagger \Sigma^{-1} x_i}{s^\dagger \Sigma^{-1} s}. \tag{S30}$$

Step 4: comparison to the OF estimator. Minimizing $\chi^2(a) = (x_i - as)^\dagger \Sigma^{-1} (x_i - as)$ over a gives

$$\hat{a}_i^{(\mathrm{OF})} = \frac{s^\dagger \Sigma^{-1} x_i}{s^\dagger \Sigma^{-1} s}, \tag{S31}$$

identical to (S30). Subspace equivalence therefore implies amplitude equivalence, which implies estimator equivalence.

Step 5: efficiency. Both estimators are unbiased with common variance

$$\mathrm{Var}(\hat{a}_i) = \frac{1}{s^\dagger \Sigma^{-1} s}, \tag{S32}$$

equal to the CRB for model (S27), so the shared estimator is efficient. □

S11.2 Proof of Theorem 4.2

For rank-1 EMPCA with basis u,

$$\hat{c} = \frac{u^\dagger \Sigma^{-1} x}{u^\dagger \Sigma^{-1} u}. \tag{S33}$$

Substituting $u = \gamma s_0$,

$$\hat{c} = \frac{\gamma^* s_0^\dagger \Sigma^{-1} x}{|\gamma|^2 s_0^\dagger \Sigma^{-1} s_0} = \frac{1}{\gamma} \frac{s_0^\dagger \Sigma^{-1} x}{s_0^\dagger \Sigma^{-1} s_0} = \frac{\hat{A}_{\mathrm{OF}}}{\gamma}, \tag{S34}$$

hence

$$\hat{s}_{\mathrm{EMPCA}} = \hat{c}\, u = \frac{\hat{A}_{\mathrm{OF}}}{\gamma}(\gamma s_0) = \hat{A}_{\mathrm{OF}}\, s_0 = \hat{s}_{\mathrm{OF}}, \tag{S35}$$

and the weighted residual $(x-\hat{s})^\dagger \Sigma^{-1} (x-\hat{s})$ is identical for both estimators. The remaining freedom $u \mapsto \alpha u$, $c \mapsto c/\alpha$ is the rank-1 gauge transformation: it changes the coefficient convention but not the reconstructed signal. □

S11.3 Extended proof of the Bridge Theorem (Theorem 3.1)

Step A: reduction to whitened form. Set $\tilde{x}_i = \Sigma^{-1/2} x_i$ and $W = \Sigma^{-1/2} P$; the constraint $P^\dagger \Sigma^{-1} P = I_k$ becomes $W^\dagger W = I_k$, and since $P P^\dagger \Sigma^{-1} x_i = \Sigma^{1/2} W W^\dagger \tilde{x}_i$, the loss (4) becomes

$$\mathcal{L}(W) = \frac{1}{N} \|\tilde{X}(I - W W^\dagger)\|_F^2. \tag{S36}$$

Step B: expansion and Rayleigh–Ritz. Expanding,

$$\mathcal{L}(W) = \frac{1}{N}\|\tilde{X}\|_F^2 - \mathrm{Tr}\big(W^\dagger \hat{R}_{\tilde{x}} W\big), \qquad \hat{R}_{\tilde{x}} = \frac{1}{N}\tilde{X}^\dagger \tilde{X} = \Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2}, \tag{S37}$$

so minimizing $\mathcal{L}$ over $W^\dagger W = I_k$ is the Rayleigh–Ritz problem maximizing $\mathrm{Tr}(W^\dagger \hat{R}_{\tilde{x}} W)$, solved by the top-$k$ eigenvectors $W^* = [v_1, \ldots, v_k]$ of $\hat{R}_{\tilde{x}}$.

Step B′: the set of minimizers. Rayleigh–Ritz gives $\mathrm{Tr}(W^\dagger \hat{R}_{\tilde{x}} W) \le \sum_{j\le k} \lambda_j$ for every $W^\dagger W = I_k$ (Ky Fan's maximum principle), with equality if and only if range(W) is an invariant subspace of $\hat{R}_{\tilde{x}}$ spanned by eigenvectors of $\lambda_1, \dots, \lambda_k$. If $\lambda_k > \lambda_{k+1}$ that subspace is unique. If $\lambda_k = \lambda_{k+1}$, any k-dimensional subspace that contains the eigenspaces of all eigenvalues strictly above λ_k and is completed inside the eigenspace of λ_k attains the bound; when $\operatorname{rank} \hat{R}_{\tilde{x}} < k$ this includes every completion inside the whitened null space $\ker \hat{R}_{\tilde{x}} = \Sigma^{1/2} \ker \hat{R}_x$, since the corresponding λ_j vanish; in detector coordinates ($P = \Sigma^{1/2} W$) these completions are the directions of $\Sigma \ker \hat{R}_x$, the null space of $\hat{R}_x \Sigma^{-1}$. This is statement 1–2 of the theorem; the feasible range $1 \le k \le d$ is the range in which $W^\dagger W = I_k$ has solutions.

Step C: translation back and connection to EMPCA. With $P^* = \Sigma^{1/2} W^*$: left-multiplying $\Sigma^{-1/2} \hat{R}_x \Sigma^{-1/2} v_j = \lambda_j v_j$ by $\Sigma^{1/2}$ gives $\hat{R}_x \Sigma^{-1} (\Sigma^{1/2} v_j) = \lambda_j (\Sigma^{1/2} v_j)$, the generalized eigenproblem whose top-k eigenspace is the EMPCA optimum (Appendix S9); every k-dimensional invariant subspace is a fixed point of the EMPCA iteration, and the top-k one is the global minimizer (Appendix S9.5). Hence, under the gap condition, range(P^*) = range$(\hat{U})$ up to a unitary rotation within the subspace.

Step D: minimum value. At the optimum, $\mathrm{Tr}(W^{*\dagger} \hat{R}_{\tilde{x}} W^*) = \sum_{j=1}^{k} \lambda_j$, so

$$\mathcal{L}^* = \frac{1}{N} \|\tilde{X}\|_F^2 - \sum_{j=1}^{k} \lambda_j = \mathrm{Tr}(\Sigma^{-1} \hat{R}_x) - \sum_{j=1}^{k} \lambda_j, \tag{S38}$$

the unexplained variance in whitened coordinates. □

Remark on uniqueness. Under the gap condition W^* is unique only up to right multiplication by $Q \in \mathcal{U}(k)$; without it the family of Step B′ applies. Physical estimators (reconstructed signals, likelihoods, χ^2 values) depend only on the projector $WW^\dagger$, which is invariant under such rotations. The population version of the theorem replaces $\hat{R}_x$ by $\mathbb{E}[xx^\dagger]$ and connects to the asymptotic regime of Theorem 4.2.

S12 Symmetry, gauge and perturbation geometry

S13 Symmetry, Gauge Freedom, and Geometric Interpretation

This appendix develops the structural background summarized in Section 3.3: the symmetry-axiomatic derivation of the noise-aware architecture, the Grassmannian geometry of the subspace family, and the finite-sample stability statement.

S13.1 Symmetry-axiomatic derivation of the noise-aware architecture

Definition S13.1 (Noise isometry group)**.**

$$G_{\text{noise}} = \{Q \in GL(\mathbb{C}^d) : Q^\dagger \Sigma^{-1} Q = \Sigma^{-1}\} \tag{S39}$$

is the group of linear isometries of the noise inner product $\langle u, v\rangle_{\Sigma^{-1}} = u^\dagger \Sigma^{-1} v$*. In whitened coordinates, every* $Q \in G_{\text{noise}}$ *becomes a unitary map* $\tilde{Q} = \Sigma^{-1/2} Q \Sigma^{1/2} \in \mathcal{U}(d)$*.*

Definition S13.2 (Latent gauge group)**.** *For a rank-k decoder basis* $P \in \mathbb{C}^{d\times k}$*, the latent gauge group is* $G_{\text{gauge}} = \mathcal{U}(k)$ *acting by*

$$P \mapsto PQ, \qquad z \mapsto Q^\dagger z, \qquad Q \in \mathcal{U}(k), \tag{S40}$$

which changes the coordinate basis of the learned subspace but leaves the reconstruction Pz *invariant. The physical object is the rank-k signal subspace and its projector, not a particular basis. This is a* global *gauge symmetry — one unitary acting on the whole latent space — not a local gauge field over a base manifold in the geometric-deep-learning sense [24].*

Theorem S13.1 (Symmetry-axiomatic uniqueness)**.** *Consider linear rank-k reconstructions* $\hat{x} = DEx$ *with* $D(z) = Pz$*,* $P^\dagger \Sigma^{-1} P = I_k$*, and quadratic residual loss* $\mathcal{L}_M(x, \hat{x}) = (x - \hat{x})^\dagger M (x - \hat{x})$*,* $M \succ 0$*. Suppose:*

1. noise-metric invariance*:* $\mathcal{L}_M(Qx, Q\hat{x}) = \mathcal{L}_M(x, \hat{x})$ *for all* $Q \in G_{\text{noise}}$*;*
2. gauge invariance*:* $P \mapsto PQ$*,* $z \mapsto Q^\dagger z$ *leaves* $\hat{x}$ *and the loss unchanged for all* $Q \in \mathcal{U}(k)$*;*
3. subspace unbiasedness*:* $DEp = p$ *for every* $p \in \text{range}(P)$*;*

4. metric-orthogonal projection*: among rank-k linear maps with range* range(P)*, DE minimizes the noise-metric residual norm.*

Then $M = \kappa\Sigma^{-1}$ for some $\kappa > 0$ (the loss is the Gaussian χ^2 up to scale); the encoder is the Σ^{-1}-adjoint of the decoder,

$$E = D^\sharp = P^\dagger\Sigma^{-1}, \qquad DE = PP^\dagger\Sigma^{-1}; \tag{S41}$$

and minimizing the resulting loss over all such rank-k subspaces yields the rank-k EMPCA subspace.

Proof. **Loss uniqueness.** Noise-metric invariance gives $Q^\dagger MQ = M$ for all $Q \in G_{\text{noise}}$. In whitened coordinates, with $\tilde{M} = \Sigma^{1/2}M\Sigma^{1/2}$ and $\tilde{Q} = \Sigma^{-1/2}Q\Sigma^{1/2}$, this reads $\tilde{Q}^\dagger\tilde{M}\tilde{Q} = \tilde{M}$ for every $\tilde{Q} \in \mathcal{U}(d)$. The only matrices commuting with all of $\mathcal{U}(d)$ are scalar multiples of the identity, so $\tilde{M} = \kappa I$ with $\kappa > 0$, i.e. $M = \kappa\Sigma^{-1}$. Equivalently, information geometry selects the same loss: for the Gaussian family with fixed covariance Σ, the Fisher information metric on the mean parameter is Σ^{-1}, and the corresponding squared geodesic distance is the Mahalanobis χ^2 distance [44].

Adjointness and gauge invariance. For $D(z) = Pz$, the adjoint with respect to $\langle\cdot,\cdot\rangle_{\Sigma^{-1}}$ in data space and the Euclidean inner product in latent space is the unique $D^\sharp$ with $\langle Dz, x\rangle_{\Sigma^{-1}} = \langle z, D^\sharp x\rangle_2$; since $\langle Pz, x\rangle_{\Sigma^{-1}} = z^\dagger P^\dagger\Sigma^{-1}x$, we get $D^\sharp x = P^\dagger\Sigma^{-1}x$. The metric-orthogonal rank-k projection onto range(P) is then $DD^\sharp = PP^\dagger\Sigma^{-1}$ using $P^\dagger\Sigma^{-1}P = I_k$. Under $P \mapsto PQ$ the encoder transforms as $Q^\dagger P^\dagger\Sigma^{-1}x$ and the reconstruction $PQQ^\dagger P^\dagger\Sigma^{-1}x = PP^\dagger\Sigma^{-1}x$ is unchanged, proving both the tied form and the gauge invariance.

EMPCA minimizer. Substituting $E = P^\dagger\Sigma^{-1}$ and $M = \kappa\Sigma^{-1}$ gives the loss (4) up to scale; expanding gives Eq. (4), and minimizing over $P^\dagger\Sigma^{-1}P = I_k$ is the Rayleigh–Ritz problem for the top-k whitened eigenspace — the rank-k EMPCA subspace. □

Corollary S13.2 (Bridge Theorem from symmetry)**.** *The noise-aware tied linear AE of Eq.* (4) *is the unique linear rank-k architecture–loss pair satisfying the requirements of Theorem S13.1, up to latent gauge rotations and an overall positive scaling of the loss; its minimizer therefore coincides with the rank-k EMPCA subspace, recovering Theorem 3.1.*

Remark S13.1 (Tied weights as the adjoint condition)**.** *The tied-weight constraint is not a regularization heuristic; it states that encoder and decoder are mutual adjoints under the correct inner products. What is lost without it is made precise next: not the optimal reconstruction, but the guarantee that intermediate reconstruction maps are Σ^{-1}-orthogonal projections at all.*

S13.2 Untied linear encoder–decoder pairs

Let $E \in \mathbb{C}^{k\times d}$ and $D \in \mathbb{C}^{d\times k}$ be unconstrained, and consider the same weighted objective

$$\mathcal{L}(D,E) = \frac{1}{N}\sum_{i=1}^{N}(x_i - DEx_i)^\dagger \Sigma^{-1}(x_i - DEx_i). \tag{S42}$$

In whitened coordinates $\tilde{D} = \Sigma^{-1/2}D$, $\tilde{E} = E\Sigma^{1/2}$, $\tilde{R} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2}$, the objective depends on the pair only through the product $B = \tilde{D}\tilde{E}$, a matrix of rank at most k:

$$\mathcal{L}(D,E) = \operatorname{tr}\big((I-B)\tilde{R}(I-B)^\dagger\big) = \big\|(I-B)\tilde{R}^{1/2}\big\|_F^2. \tag{S43}$$

Proposition S13.3 (Untied optimum)**.** *Let $\tilde{R} = V\Lambda V^\dagger$ with $\lambda_1 \geq \cdots \geq \lambda_d$. (i) The minimum of* (S43) *over all B of rank at most k equals $\sum_{j>k}\lambda_j$, the tied minimum of Theorem 3.1, and is attained at $B = V_kV_k^\dagger$, the orthogonal projector onto a top-k eigenspace. (ii) If $\tilde{R} \succ 0$ (the whitened data have full rank, which requires $N \geq d$) and $\lambda_k > \lambda_{k+1}$, this minimizer is unique: every optimal untied pair has $DE = P^*P^{*\dagger}\Sigma^{-1}$ with P^* the tied optimum, so the untied optimum is the Σ^{-1}-orthogonal projection and tied and untied classes have the same optimal reconstruction map. (iii) If $\tilde{R}$ is singular, B is determined only on* range $\tilde{R}$ *and may act arbitrarily on its null space; if $\lambda_k = \lambda_{k+1}$ the eigenspace is not unique, as in Theorem 3.1. (iv) The parameterization is redundant: $(\tilde{D},\tilde{E}) \mapsto (\tilde{D}S, S^{-1}\tilde{E})$ for any invertible $S \in GL(k)$ leaves B and the objective unchanged, a k^2-dimensional gauge that the tie $E = D^\sharp$ reduces to the $\mathcal{U}(k)$ of Definition S13.2.*

Proof. When $\tilde{R} \succ 0$, $\{B\tilde{R}^{1/2} : \operatorname{rank} B \leq k\}$ is the set of all matrices of rank at most k, so (i)–(ii) are the Eckart–Young theorem applied to $\tilde{R}^{1/2} = V\Lambda^{1/2}V^\dagger$: the best rank-$k$ approximation is $V_k\Lambda_k^{1/2}V_k^\dagger$, unique under the gap condition, and $B = V_k\Lambda_k^{1/2}V_k^\dagger\tilde{R}^{-1/2} = V_kV_k^\dagger$. Unwhitening gives $DE = \Sigma^{1/2}V_kV_k^\dagger\Sigma^{-1/2} = P^*P^{*\dagger}\Sigma^{-1}$ with $P^* = \Sigma^{1/2}V_k$. When $\tilde{R}$ is singular, $B\tilde{R}^{1/2}$ ranges only over matrices whose row space lies in range $\tilde{R}$, which determines B on that range and not on its complement, giving (iii). Statement (iv) is immediate from $B = \tilde{D}\tilde{E}$. □

The proposition fixes what may and may not be said about untying. It may *not* be said that an untied linear autoencoder trained under the Σ^{-1} loss has a different optimal reconstruction from the tied one: under full-rank data and a spectral gap the optima coincide. What is true is that away from the optimum the product DE of an untied pair is in general not idempotent and not Σ^{-1}-self-adjoint — it is a non-self-adjoint reconstruction map, not a projection — and that the optimizer must traverse the $GL(k)$ gauge of statement (iv), whose flat directions an anisotropic Σ can make badly conditioned for a first-order method. Whether a first-order optimizer traverses that gauge in practice is an empirical question; the gauge-conditioning explanation is a plausible mechanism, not a proved one.

Remark S13.2 (Connection to geometric deep learning)**.** *In geometric-deep-learning terms [24], G_{noise} is the symmetry group of the observation-space metric induced by the noise model, and $G_{\text{gauge}} = \mathcal{U}(k)$ is the global gauge freedom of the latent basis. The blueprint usually derives architectures from symmetries of the input domain; here the symmetry is information-geometric, imposed by the detector covariance. The Bridge Theorem is thus not an algebraic coincidence but the linear reconstruction method selected by the noise metric and the latent gauge freedom.*

S13.3 Grassmannian geometry of the subspace family

When $\tilde{U}^\dagger \tilde{U} = I_k$, the EMPCA objective depends on $\tilde{U}$ only through the orthogonal projector $\Pi_{\tilde{U}} = \tilde{U}\tilde{U}^\dagger$: EMPCA optimizes not over bases but over the space of k-dimensional subspaces of $\mathbb{C}^d$, the complex Grassmannian

$$\mathrm{Gr}(k, d) = \mathcal{U}(d)/(\mathcal{U}(k) \times \mathcal{U}(d-k)), \tag{S44}$$

equivalently the space of rank-k orthogonal projectors in whitened coordinates. The $\mathcal{U}(k)$ gauge freedom is precisely the freedom to choose a basis within a fixed point of $\mathrm{Gr}(k, d)$. For $k = 1$, $\mathrm{Gr}(1, d) = \mathbb{C}P^{d-1}$ and basic OF corresponds to the fixed point represented by the whitened template direction $[\tilde{s}_0]$. This places EMPCA within the standard geometry of Grassmannian subspace optimization [29, 30].

S13.4 Product gauge of the separable class

For the NFPA class of Section 4.4 the reconstruction depends on the factors only through the projectors $U_c U_c^T$ and $U_t U_t^T$, so the gauge group is the product $O(k_c) \times O(k_t)$ acting by $U_c \mapsto U_c Q_c$, $U_t \mapsto U_t Q_t$. Under the embedding $\mathrm{range}(U_t) \otimes \mathrm{range}(U_c)$ this is the subgroup $\{Q_t \otimes Q_c\}$ of the $O(k_c k_t)$ gauge of the unrestricted rank-$k_c k_t$ class: separability removes not only most of the subspaces but also most of the basis freedom within a subspace. The practical rule is unchanged — interpret projectors, subspaces, reconstructions, and principal angles, never raw factor coordinates — and it now applies axis by axis. In particular a comparison of a learned factor with a planted one must be made between the projectors $U_c U_c^T$ and $V_c V_c^T$ (or between subspaces), which is gauge-invariant, and not between the matrices U_c and V_c, which is not.

Proof of Proposition 3.3. The identity (6) telescopes. $a \geq 0$ because $\mathcal{F} \subset \mathrm{Gr}(k, Cd)$ and $P_U^\star$ minimizes R_K over all rank-k orthogonal projectors (Ky Fan's maximum principle: $\mathrm{tr}(PK) \leq \sum_{i \leq k} \mu_i$); $x_{\mathcal{F}}, x_U \geq 0$ by the definition of the two optima. For the first inequality of (7) write $w_i = v_i^T P_F v_i \in [0, 1]$, so that $\sum_i w_i = \mathrm{tr}\, P_F = k$ and $\mathrm{tr}(P_F K) = \sum_i \mu_i w_i$.

Then

$$\begin{aligned}\operatorname{tr}(P_F K) &\le \sum_{i\le k}\mu_i w_i + \mu_{k+1}\sum_{i>k} w_i = \sum_{i\le k}\mu_i - \sum_{i\le k}(\mu_i-\mu_{k+1})(1-w_i)\\ &\le \operatorname{tr}(P_U^\star K) - \Delta_k \sum_{i\le k}(1-w_i),\end{aligned}$$

and $\sum_{i\le k}(1-w_i) = k - \operatorname{tr}(P_F P_U^\star)$. For the second, $P_F = P_t \otimes P_c \preceq I \otimes P_c$ and $P_F \preceq P_t \otimes I$ in the Loewner order, and $P_U^\star \succeq 0$, so $\operatorname{tr}(P_F P_U^\star) \le \operatorname{tr}\big((I\otimes P_c)P_U^\star\big) = \operatorname{tr}(P_c M_c)$ with $M_c = \operatorname{tr}_t P_U^\star$ the partial trace over the temporal factor, a positive semidefinite $C\times C$ matrix of trace k; over rank-k_c projectors $\operatorname{tr}(P_c M_c)$ is at most the sum of the k_c largest eigenvalues of M_c (Ky Fan again), and likewise $\operatorname{tr}(P_F P_U^\star) \le \sum_{i\le k_t}\mu_i(M_t)$ with $M_t = \operatorname{tr}_c P_U^\star$; the cap k is $\operatorname{tr} P_F$. Finally, if $\Delta_k > 0$ and $a = 0$ then some $P_F \in \mathcal{F}$ attains $R_K(P_U^\star)$, so $\operatorname{tr}(P_F P_U^\star) = k$ by the first inequality, and two rank-k orthogonal projectors with $\operatorname{tr}(P_F P_U^\star) = k$ are equal; conversely $P_U^\star \in \mathcal{F}$ gives $a = 0$ directly. The bound is computed on whatever K is supplied; in Section 5.3 that is an estimate of the generator's clean-signal covariance from 2×10^5 events, so the numbers there carry that estimate's error and floating-point error, and nothing more. □

S13.5 Metric symmetry versus domain symmetry

It is essential not to conflate the metric symmetry G_{noise} with a domain symmetry. G_{noise} preserves the residual norm; it acts on observation space and knows nothing about the detector layout. It is not time translation, channel relabeling, detector rotation, or a graph automorphism — transformations that act on the measurement domain and may (approximately) commute with the physical signal-generating process, and from which the geometric-deep-learning blueprint derives its architectures [24]. Prewhitening therefore fixes the residual metric but does not by itself impose convolutional locality, graph equivariance, or deformation stability; those inductive biases must be encoded by the architecture, and their validity is a property of the detector, not of the noise model. The same distinction separates the two symmetry groups of Theorem S13.1: G_{noise} is information-geometric and selects the loss; G_{gauge} is a global redundancy of the parameterization and selects nothing. Neither is a domain symmetry, and the separable prior of Section 4.4 is the first place in the hierarchy where a domain assumption — that channel and time structure factor — enters at all.

S13.6 Representation geometry: proofs of Propositions 2.4 and 2.4

Both proofs are for real observations, real latents and real Gaussian noise, as the propositions are stated.

Proof of Proposition 2.4. The log-likelihood of x given ϑ is $\ell(\vartheta) = -\frac{1}{2}(x - g(\vartheta))^T \Sigma^{-1}(x - g(\vartheta)) + \text{const}$, with score $\nabla_\vartheta \ell = J_g(\vartheta)^T \Sigma^{-1}(x - g(\vartheta))$. Since $\mathbb{E}[(x - g(\vartheta))(x - g(\vartheta))^T] = \Sigma$, the Fisher information is $\mathbb{E}[\nabla_\vartheta \ell \, \nabla_\vartheta \ell^T] = J_g^T \Sigma^{-1} \Sigma \Sigma^{-1} J_g = J_g^T \Sigma^{-1} J_g$, which is positive semidefinite, and positive definite exactly when $J_g v = 0$ has only the trivial solution. For the representation statement, compose $h \mapsto y = g_{\text{out}}(h) \in \mathbb{R}^d$ with the observation model: a displacement δh moves the predicted observation by $J_h \delta h$ to first order, whose squared whitened length is $\delta h^T J_h^T \Sigma^{-1} J_h \delta h$; the form is degenerate on $\ker J_h$. The construction requires y to live in observation space, because it is Σ that supplies the inner product. The identifiability statement is the first-order one: if $J_g(\vartheta) v = 0$ then $g(\vartheta + \epsilon v) = g(\vartheta) + O(\epsilon^2)$, so no estimator distinguishes ϑ from $\vartheta + \epsilon v$ at rate ϵ; this does not exclude that g is injective near ϑ. Throughout, g is assumed continuously differentiable at the point of linearization. □

Proof of Proposition 2.4. Under (iii) the reconstruction acts on $u = t + w$ with $t \in T_S$, $\mathbb{E}[t] = 0$ by (iv), $\mathbb{E}[tw^T] = 0$ by (ii), and $\mathbb{E}[ww^T] = I$ by (i), so $\mathbb{E}[uu^T] = R_t + I$. For a rank-$k$ orthogonal projector Π, $J(\Pi) = \operatorname{tr}\big((I - \Pi)(R_t + I)(I - \Pi)\big) = \operatorname{tr}\big((I - \Pi)R_t\big) + \operatorname{tr}(I - \Pi) = \operatorname{tr} R_t - \operatorname{tr}(\Pi R_t) + (d - k)$, using $(I - \Pi)^2 = I - \Pi$ and the cyclicity of the trace; this is statement 1. If $Q \in O(d)$ fixes T_S pointwise then, because $\operatorname{range} R_t \subseteq T_S$, $QR_tQ^T = R_t$ and $\operatorname{tr}(Q\Pi Q^T R_t) = \operatorname{tr}(\Pi \, Q^T R_t Q) = \operatorname{tr}(\Pi R_t)$, so $J(Q\Pi Q^T) = J(\Pi)$ exactly. For statement 2, $\operatorname{tr}(\Pi R_t) \le \sum_{j \le k} \mu_j$ with μ_j the eigenvalues of R_t (Ky Fan), with equality if and only if $\operatorname{range} \Pi$ contains a top-k eigenspace of R_t; when $k \ge r = \operatorname{rank} R_t$ the top-k eigenspaces are exactly the k-dimensional subspaces containing $\operatorname{range} R_t$, whose remaining $k - r$ dimensions are arbitrary in $(\operatorname{range} R_t)^\perp$, and when $k < r$ the maximizers are the top-k eigenspaces of R_t with the usual gap condition. For statement 3, with $B = \Sigma^{-1/2} A_{\cdot S}$ and $K \succeq 0$, $\operatorname{range}(BKB^T) = \operatorname{range}(BK^{1/2}) = B(\operatorname{range} K)$, so $\operatorname{range} R_t = \operatorname{range} B = T_S$ if and only if $B(\operatorname{range} K) = B(\mathbb{R}^{|S|})$, i.e. $\operatorname{range} K + \ker B = \mathbb{R}^{|S|}$; taking orthogonal complements, $\ker K \cap (\ker B)^\perp = \{0\}$, which is positive definiteness of K on $\operatorname{range}(B^T)$. If $K \succ 0$ the condition holds for every B; if B is injective, $\ker B = \{0\}$ and the condition reduces to $K \succ 0$. The condition is not symmetric in the two factors: $B = [1 \; 1]$ with $K = \operatorname{diag}(1, 0)$ gives $BKB^T = 1$ and $\operatorname{range} R_t = \operatorname{range} B$ although K is singular and B is not injective. □

The proof makes visible why the finite-sample situation is different. Replacing $R_t + I$ by the sample second moment $\hat{R}_u = N^{-1} \sum_i u_i u_i^T$ replaces the isotropic term I by a sample

noise covariance and R_t by a finite-sample estimate. For Gaussian w and $d/N \to \gamma$, the nonzero eigenvalues of the sample noise covariance follow the Marchenko–Pastur law on $[(1-\sqrt{\gamma})^2, (1+\sqrt{\gamma})^2]$, and when $N < d$ there are in addition $d-N$ eigenvalues exactly at zero [52]. The population objective's indifference to the complement then becomes a small but nonzero preference set by the noise realization. For a single population direction of excess variance θ added to isotropic noise, the leading sample eigenvalue separates from the bulk, and the leading sample eigenvector acquires a nonvanishing asymptotic overlap with that direction, if and only if $\theta > \sqrt{\gamma}$; below the transition the overlap vanishes, and above it equals $\big((1-\gamma/\theta^2)/(1+\gamma/\theta)\big)^{1/2}$, not one [33, 53]. These are statements about an isolated spike, isotropic noise and unstructured PCA; for several planted directions and for the structured NFPA estimator they are used in the controlled evidence (the controlled separable-class evidence, held from this version) only as diagnostics, not as a threshold.

S13.7 Finite-sample stability: extended Davis–Kahan discussion

Two perturbations act on the whitened operator whose top-k eigenspace the estimators of this paper return, and both are handled by the Davis–Kahan theorem in the same coordinates. Throughout, $f(M)$ maps a Hermitian operator to its top-k eigenspace, a point of $\mathrm{Gr}(k,d)$, $\sin\Theta$ is the sine of the largest principal angle (equivalently the operator-norm distance between the two orthogonal projectors, which is a metric on $\mathrm{Gr}(k,d)$), and the form of the theorem used is that of Yu, Wang and Samworth [48]: for Hermitian M and $\hat{M}$ with $\Delta_\lambda = \lambda_k(M) - \lambda_{k+1}(M) > 0$, $\sin\Theta(f(\hat{M}), f(M)) \le 2\|\hat{M} - M\|_{\mathrm{op}}/\Delta_\lambda$.

Sampling error at known covariance. With the true Σ, $A = \Sigma^{-1/2} R_x \Sigma^{-1/2}$ and $\hat{A} = \Sigma^{-1/2}\hat{R}_x\Sigma^{-1/2} = A + \Sigma^{-1/2}\Delta R\,\Sigma^{-1/2}$, so

$$\sin\Theta\big(f(\hat{A}), f(A)\big) \le \frac{2\,\|\Sigma^{-1/2}\Delta R\,\Sigma^{-1/2}\|_{\mathrm{op}}}{\Delta_\lambda}, \tag{S45}$$

which is (5): numerator and gap are both whitened. The numerator is a covariance-estimation error and is controlled under the tail and sample-size assumptions stated in Section 4.2; an unqualified $O(\sqrt{d/N})$ is not claimed.

Covariance error (proof of the bound in Remark 3.4). Let $\hat{\Sigma} = \Sigma + \Delta\Sigma$ with $E = \Sigma^{-1/2}\Delta\Sigma\,\Sigma^{-1/2}$, $\varepsilon = \|E\|_{\mathrm{op}} \le \frac{1}{2}$, so that $\hat{\Sigma} = \Sigma^{1/2}(I+E)\Sigma^{1/2}$ and $I + E \succ 0$. A basis $\hat{P}$ fitted with $\hat{\Sigma}$ satisfies the generalized eigenproblem $R_x\hat{\Sigma}^{-1}\hat{p} = \lambda\hat{p}$ (Theorem 3.1, item 1, with $\hat{\Sigma}$ in place of Σ). Substituting $\hat{p} = \Sigma^{1/2}u$ gives, in the true whitened coordinates, $A(I+E)^{-1}u = \lambda u$, and $A(I+E)^{-1} = (I+E)^{1/2}\tilde{A}\,(I+E)^{-1/2}$ with the Hermitian operator

$\tilde{A} = (I+E)^{-1/2} A\,(I+E)^{-1/2}$. Hence $\hat{\mathcal{S}} = \mathrm{range}(\Sigma^{-1/2}\hat{P}) = (I+E)^{1/2} f(\tilde{A})$, while $\mathcal{S} = f(A)$. Two steps bound the distance. (i) With $G = (I+E)^{-1/2} - I$, $\|G\| \le (1-\varepsilon)^{-1/2} - 1 \le \varepsilon$ for $\varepsilon \le \frac{1}{2}$, and $\tilde{A} - A = GA + AG + GAG$, so $\|\tilde{A} - A\| \le (2\varepsilon + \varepsilon^2)\lambda_1(A) \le \frac{5}{2}\varepsilon\lambda_1(A)$ and $\sin\Theta(f(\tilde{A}), f(A)) \le 5\varepsilon\lambda_1(A)/\Delta_\lambda$. (ii) With $H = (I+E)^{1/2} - I$, $\|H\| \le \varepsilon/(1+\sqrt{1-\varepsilon})$, and for any k-dimensional subspace Z and unit $z \in Z$ the vector $(I+H)z$ lies within $\|H\|$ of Z and has norm at least $1 - \|H\|$, so $\sin\Theta((I+H)Z, Z) \le \|H\|/(1-\|H\|) \le \varepsilon$ for $\varepsilon \le \frac{1}{2}$. The triangle inequality for $\sin\Theta$ then gives $\sin\Theta(\hat{\mathcal{S}}, \mathcal{S}) \le 5\varepsilon\lambda_1/\Delta_\lambda + \varepsilon \le 6\varepsilon\lambda_1/\Delta_\lambda$, which is (5). The bound is invariant under a common scaling of Σ, $\Delta\Sigma$ and R_x, as it must be; the regression check `scripts/check_revision_claims.py` verifies it on random instances and reproduces the scaling counterexample that rules out an absolute bound in $\|\Delta\Sigma\|$. If the fitted bases are compared in detector coordinates, the comparison must be made in the Σ^{-1} metric, which is what the whitened statement does; a Euclidean comparison in detector coordinates additionally depends on the changing unwhitening map $\hat{\Sigma}^{1/2}$ and is not the quantity bounded here. When both perturbations act at once — a sample second moment $\hat{A} = A + \Delta A$ whitened with an estimated covariance — the operator whose eigenspace is fitted is, in true whitened coordinates and up to the same transport, $(I+E)^{-1/2}(A+\Delta A)(I+E)^{-1/2} = \tilde{A} + (I+E)^{-1/2}\Delta A(I+E)^{-1/2}$; the sample perturbation is conjugated, not added unchanged, and $\|(I+E)^{-1/2}\Delta A(I+E)^{-1/2}\| \le \|\Delta A\|/(1-\varepsilon)$. Hence, under the same hypotheses, $\sin\Theta(\hat{\mathcal{S}}, \mathcal{S}) \le 2\big[(2\varepsilon+\varepsilon^2)\lambda_1(A) + \|\Delta A\|/(1-\varepsilon)\big]/\Delta_\lambda + \varepsilon$, by the same two steps.

First-order displacement (proof of Proposition 3.2). In the true whitened coordinates the basis fitted with $\hat{\Sigma}$ spans the top-k invariant subspace of $A(I+E)^{-1}$ (previous paragraph). For $\varepsilon < 1$, $(I+E)^{-1} = I - E + O(\varepsilon^2)$, so $A(I+E)^{-1} = A + M + O(\varepsilon^2)$ with the non-symmetric perturbation $M = -AE$ of the symmetric operator A, $\|M\| \le \varepsilon\lambda_1(A)$. First-order perturbation of a simple invariant subspace [49, Ch. V] applies: with $Q_{\mathcal{R}} = [q_1, \ldots, q_k]$ and $Q_{\mathcal{R}^\perp}$ its complement, the top-k invariant subspace of $A + M$ is spanned by the columns of $Q_{\mathcal{R}} + Q_{\mathcal{R}^\perp} X$ with $X_{ji} = M_{ji}/(\lambda_i - \lambda_j) + O(\|M\|^2/\delta^2)$ for $i \in \mathcal{R}$, $j \notin \mathcal{R}$, where $M_{ji} = q_j^T M q_i = -\lambda_j E_{ji}$ and $\delta = \min|\lambda_i - \lambda_j|$ over such pairs. The orthogonal projector onto that subspace is $P + Q_{\mathcal{R}^\perp} X Q_{\mathcal{R}}^T + Q_{\mathcal{R}} X^T Q_{\mathcal{R}^\perp}^T + O(\|X\|^2)$, which is (5). The blocks $M_{\mathcal{R}\mathcal{R}}$ and $M_{\mathcal{R}^\perp\mathcal{R}^\perp}$ enter X only at second order — they change the eigenbasis inside $\mathcal{R}$ and inside $\mathcal{R}^\perp$, not the two subspaces — which is item 1. Item 2 reads off the Frobenius norm of the two off-diagonal blocks and uses $\sin\Theta(\hat{\mathcal{S}}, \mathcal{S}) = \|(I-P)\hat{P}\|_{\mathrm{op}} = \|Q_{\mathcal{R}^\perp} X Q_{\mathcal{R}}^T\|_{\mathrm{op}} + O(\|X\|^2)$. For item 3, if $E = \sum_i e_i q_i q_i^T$ then $A(I+E)^{-1} = \sum_i \lambda_i (1+e_i)^{-1} q_i q_i^T$ exactly; its eigenvectors are the q_i, and its top-k eigenvectors are those of $\mathcal{R}$ if and only if the stated inequality holds. For item 4, replace A by $A + D$ before expanding: $(A+D)(I+E)^{-1} = A + (D - AE) + O(\varepsilon^2, \varepsilon\|D\|)$, so $M_{ji} = D_{ji} - \lambda_j E_{ji}$ in the eigenbasis of A; expanding about $\hat{A} = A + D$ in its own eigenbasis

instead, $M = -\hat{A}E$ and D does not appear. The remainder statement is the second-order term of the same expansion, $\|X\| \le \|M\|/\delta + O(\|M\|^2/\delta^2)$, which with $\|M\| \le \varepsilon\lambda_1(A)$ is $O(\varepsilon^2\lambda_1^2/\delta^2)$. Because $\delta \le \lambda_k \le \lambda_1$, this dominates the $O(\varepsilon^2\lambda_1/\delta)$ contributed by truncating $(I+E)^{-1} = I - E + O(\varepsilon^2)$, and it is the order the proposition states: the expansion is informative when $\varepsilon\lambda_1/\delta \ll 1$ rather than merely when $\varepsilon \ll 1$. (An earlier statement of the proposition gave the remainder as $O(\varepsilon^2\lambda_1/\delta)$; that is violated as soon as λ_1/δ exceeds one, as `scripts/check_revision_claims.py` exhibits on a coupling-block perturbation.) Note that the transported subspace $(I+E)^{1/2}f(\tilde{A})$ of the previous paragraph and the invariant subspace of $A(I+E)^{-1}$ used here are the same object; the proposition expands it directly rather than through the symmetrized $\tilde{A}$, which is why the within-block entries of E drop out exactly at first order.

Reading. Structurally the bound plays the role of a deformation-stability statement in geometric deep learning [24]: small perturbations of the estimated object produce small changes in the geometric estimator when the stability constant — here the conditioning ratio λ_1/Δ_λ of the retained whitened spectrum — is controlled. The relevant perturbation is a *signal-space* deformation, the covariance estimate changing on the fixed observation domain, as opposed to a *domain* deformation as in graph or manifold settings. Experiment S7 (Section 5) is the empirical counterpart; the CRESST tied-AE row is a closed-form parameterization check (Section 6.2) and carries no information about stability. For the separable class the analogous statement would bound the angle between the NFPA and EMPCA subspaces by the best Tucker-2 approximation error to the EMPCA projector [20] divided by the spectral gap; Section 3.5 states this as an open target.

S14 Extended experiments and deferred diagnostics

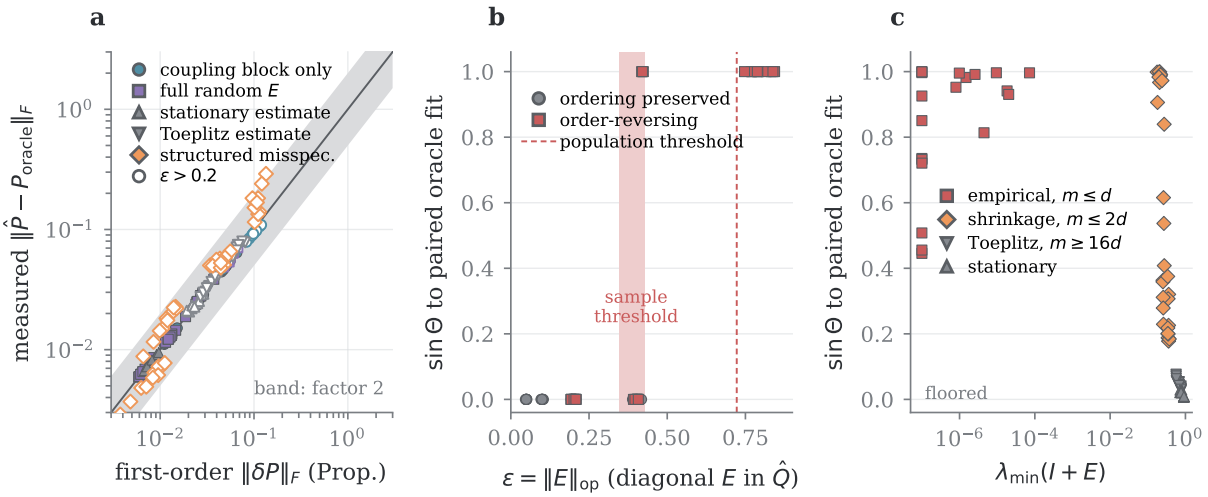


Figure S1: Covariance-perturbation anatomy (T19; ten seeds per arm, $d = 256$, $k = 2$, $N = 512$; displacement measured against the paired oracle fit on the same training draw). (a) Measured Frobenius displacement of the fitted projector against the first-order value of Proposition 3.2 for perturbations built in the sample eigenbasis: a pure coupling block (C), a full random symmetric E (F), and estimated/misspecified covariances (M); filled markers are the primary band $\varepsilon \leq 0.2$, the band is a factor of two about the identity. (b) Diagonal E: ordering-preserving perturbations (P) leave the subspace unchanged at every ε tested; order-reversing ones (R) exchange it exactly at the sample-spectrum reversal (shaded: per-seed sample thresholds; dashed: the population value). (c) Conditioning: subspace error against $\lambda_{\min}(I + E)$ for the near-singular empirical covariances (consistent decoder, v2), the shrinkage estimates, and the well-conditioned arms. Registered inputs and field names: `generated/SOURCE_MAP_part2.md`.

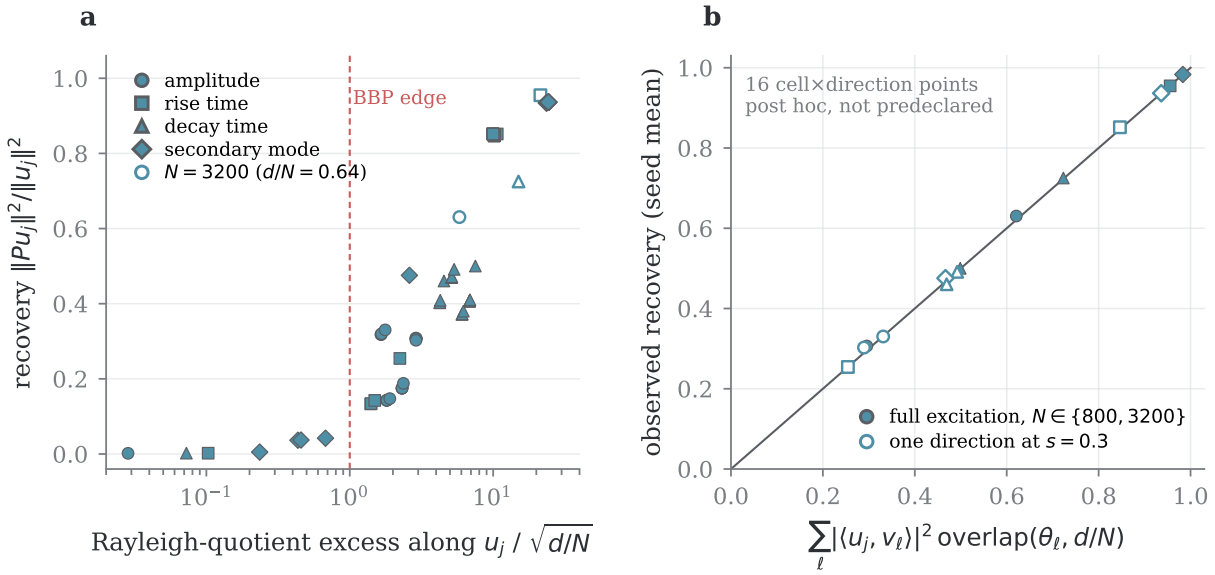


Figure S2: Excitation and detectability (T21; planted family at $\alpha = 0$, $C = 8$, $T = 256$, rank four, oracle covariance, ten seeds per cell). (a) Seed-mean recovery of each planted direction against the Rayleigh quotient of the training-distribution signal covariance along it, in excess of the noise floor and relative to $\sqrt{d/N}$; recovery is not a function of this scalar. (b) *Post hoc, not predeclared*: recovery predicted by the isolated-spike overlap weighted by each direction's squared eigenvector loadings, against the observed seed mean. Agreement is within 0.015 on every point; it is a description of these runs, and its test on fresh data is left open. Registered inputs: `generated/SOURCE_MAP_part2.md`.

S15 Supplementary Evidence Status

Earlier drafts included quantitative panels for principal-component interpretation, Cramér–Rao checks, resolution scaling, template mismatch, time-shift recovery, rank saturation, timing-bias reduction, residual calibration, and iterative convergence. The current evidence bundle does not map those displayed values to exact configurations, seeds, machine-readable metrics, and a code revision. In particular, it does not independently substantiate the repeated residual values 1029.2, 1028.9, and 1028.6, the reported scaling slope, the stated timing-bias reduction, or the claimed Kolmogorov–Smirnov transition. Those quantitative claims and their legacy figures are therefore omitted from this revision rather than treated as verified results. The controlled S1–S9 records reported in Section 5 remain the synthetic evidence for the claims made in the main text.

A conditional noise-budget attribution analysis, which would have compared the discrete optimal-filter information computed from a baseline total PSD against the information obtained after removing or rescaling individual additive components, was also prepared for an earlier draft. It is not reported here. Such quantities are conditional attributions under a declared additive model: they do not show that a component can be physically removed, that its label is uniquely identifiable from a total spectrum, or that hardware performance would improve by the corresponding amount. Reporting one would additionally require the sampling frequency, record length, template normalization, endpoint factors, frequency support, and acquisition transfer function to be frozen and independently verified, which the available preparation record does not establish. The claim is therefore withdrawn from this revision rather than reported conditionally.

S16 Simulation and sampling contract

S17 Noise Sampling and Spectral Conventions

This appendix fixes the discrete spectral conventions used by the synthetic experiments of Section 5 and referenced by the optimal-filter derivations of Appendix S5. It concerns sampling and normalization only; no detector, instrument, or measured noise budget is described here.

S17.1 Generating colored noise from a one-sided PSD

The synthetic noise used throughout is a real Gaussian time-domain trace drawn from a specified one-sided PSD density $P(f)$. For trace length N, sampling frequency f_s, spacing $\Delta f = f_s/N$, and rFFT bins $f_k = k\Delta f$, we adopt the NumPy convention of an unnormalised forward transform with a factor $1/N$ in `irfft`. For each interior positive-frequency bin we draw

$$z_k = \frac{a_k + ib_k}{\sqrt{2}}, \qquad a_k, b_k \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \qquad \tilde{\eta}_k = \sqrt{\frac{f_s N P_k}{2}}\, z_k, \tag{S46}$$

where $P_k = P(f_k)$, so that $\mathbb{E}|\tilde{\eta}_k|^2 = f_s N P_k/2$. The DC coefficient and, for even N, the Nyquist coefficient have no independent imaginary part and are drawn as real Gaussians,

$$\tilde{\eta}_k = \sqrt{f_s N P_k}\, a_k, \qquad \mathbb{E}|\tilde{\eta}_k|^2 = f_s N P_k. \tag{S47}$$

For odd N, the highest positive-frequency bin remains an interior complex bin. With this convention, Parseval's identity gives the discrete one-sided relation

$$\mathbb{E}\left[\frac{1}{N}\sum_{n=0}^{N-1}\eta_n^2\right] = \Delta f \sum_{k=0}^{\lfloor N/2\rfloor} P_k, \tag{S48}$$

which is the identity used to convert between DFT-normalized quantities and physical one-sided PSD units in the optimal-filter expressions. Conventions in the literature differ by a factor of two between one-sided and two-sided PSDs, and by a further factor of two between real-valued and complex Gaussian models; the two-factor combination is held fixed across all methods within each comparison so that no ordering reported in this paper depends on it. The stochastic DC component is set to zero and any deterministic mean is represented separately. Built-in spectral shapes and custom tabular PSDs use the same density-to-bin-power conversion.

S17.2 Sampling contract and its verification

Drawing independent coefficients on a finite DFT grid defines a periodic stationary process whose covariance is circulant, not Toeplitz. When the requested covariance is a genuinely non-periodic Toeplitz matrix, a separate exact finite-Toeplitz sampler is used instead; these are distinct sampling contracts and are not interchangeable at finite N. The distinction matters for the residual-calibration test S8, where the whitened residual is compared against its nominal χ^2 distribution.

Figure S3 is a second-order spectral check only: agreement of the ensemble mean spectrum with the target density verifies Eqs. (S46)–(S48) but does not by itself establish Gaussianity. Fourier-bin normality, periodogram variability, endpoint scaling, and finite-grid covariance agreement are tested separately in the generator validation suite and are not reproduced here.

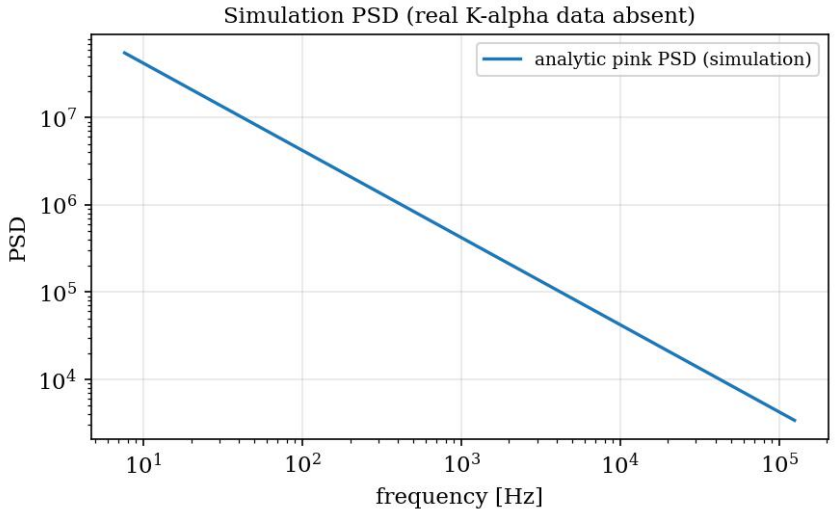


Figure S3: Second-order PSD audit for the one-sided convention of Eq. (S46) used by the synthetic OF and EMPCA experiments. Agreement of the ensemble mean spectrum with the target does not by itself establish Gaussianity; Fourier-bin normality, periodogram variability, endpoint scaling, and finite-grid covariance are tested separately.

S18 Symbols

S19 List of Symbols

Notation used throughout the paper. Complex notation with † (conjugate transpose) is used by default; all statements specialize to real data with $\dagger \to T$. Vectors are columns; whitened quantities carry tildes.

Category	Symbol	Meaning
Dimensions	d	Trace length (samples or frequency bins per event).
	N	Number of events in the dataset.
	k	Rank of the learned signal subspace (also frequency-bin index; clear from context).
	C	Number of detector channels.
Indices	i	Event index, $i = 1, \ldots, N$.
	k	Frequency-bin index of the DFT.
	c	Channel index, $c = 1, \ldots, C$.
	r	EM iteration index.
Data	$x,\ x_i$	Observed trace (column vector in $\mathbb{C}^d$).
	X_k	Frequency-domain coefficient of the observed trace at bin k.
	X	Data matrix, rows $x_i^\dagger$ ($N \times d$).
	$\tilde{x},\ \tilde{X}$	Whitened trace / data matrix, $\tilde{x} = \Sigma^{-1/2} x$.
Signal model	$s_0,\ S_k$	Fixed signal template (time / frequency domain).
	$\hat{s}$	Reconstructed signal.
	$A,\ a_i$	Signal amplitude (pulse height) of an event.
	t_0	Signal arrival time (time offset).
	U	Learned rank-k basis, $U \in \mathbb{C}^{d \times k}$, $U^\dagger \Sigma^{-1} U = I_k$.
	$c,\ \Gamma$	Latent coefficient vector / matrix $\Gamma = [c_1, \ldots, c_N]$ for the rank-k model.

Category	Symbol	Meaning
	P	Decoder basis of the noise-aware linear AE, $P^\dagger \Sigma^{-1} P = I_k$.
	z	Latent code of the autoencoder, $z = P^\dagger \Sigma^{-1} x$.
	$E,\ D$	Encoder / decoder of a linear autoencoder; $E = D^\sharp$ in the tied form.
	W	Whitened decoder basis $W = \Sigma^{-1/2} P$, $W^\dagger W = I_k$ (Appendix S11.3).
	$\omega_{kj},\ \Omega$	Per-element weights of the frequency-domain EMPCA variant (Appendix S9.3).
Noise	$n,\ N_k$	Additive noise (time / frequency domain).
	Σ	Noise covariance matrix ($\succ 0$); diagonal in frequency space for stationary noise.
	Σ^{-1}	Inverse covariance; the observation-space metric.
	$\Sigma^{-1/2}$	Whitening transform.
	$J(f),\ J_k$	Noise power spectral density (PSD).
Optimal filtering	χ^2	Mahalanobis / weighted least-squares objective, Eq. (2).
	$\Phi_k,\ \Phi_k^{(0)}$	Unnormalized / normalized OF kernels.
	N_Φ	Filter normalization $\sum_k \lvert S_k\rvert^2 / J_k$; total SNR budget.
	$\sigma_A,\ \sigma_E$	Amplitude / energy resolution.
	$B(t_0)$	Delayed matched-filter response (time-shift OF).
Subspace estimation	$\hat{R}_x$	Sample covariance $N^{-1} \sum_i x_i x_i^\dagger$.
	λ_j	Eigenvalues of $\Sigma^{-1} \hat{R}_x$ (descending).
	Δ_λ	Spectral gap between signal and noise eigenvalues.
	$\rho_w,\ \theta_w$	Weighted (whitened-space) subspace cosine / angle.

Category	Symbol	Meaning
	$\Theta(\cdot,\cdot)$	Vector of principal angles between two subspaces; $\sin\Theta$ denotes the corresponding sine (Remark 3.4).
	$\hat{\Sigma}$	Covariance estimated from calibration data, used in place of Σ.
	$\Delta\Sigma$	Covariance estimation error, $\Delta\Sigma = \hat{\Sigma} - \Sigma$.
	ΔR	Sampling error of the covariance, $\hat{R}_x = R_x + \Delta R$ (Appendix S13.7).
	μ	Ridge parameter of the regularized coefficient solve, Eq. (S25).
	$P_{\tilde{U}}$	Orthogonal projector onto the whitened subspace.
	$\mathrm{Gr}(k,d)$	Grassmannian of k-dimensional subspaces.
Separable class	$Z,\ \tilde{Z}$	Two-axis (channel $\times$ time) event matrix in $\mathbb{R}^{C\times d}$ and its per-axis whitened form.
	$\Sigma_c,\ \Sigma_t$	Channel and temporal factors of a separable covariance, $\Sigma = \Sigma_t \otimes \Sigma_c$.
	$U_c,\ U_t$	Orthonormal channel / temporal factor bases of the NFPA, $U_c \in \mathbb{R}^{C\times k_c}$, $U_t \in \mathbb{R}^{d\times k_t}$.
	$k_c,\ k_t$	Channel / temporal factor ranks; total rank $k_c k_t$.
	Γ_i	NFPA code matrix $U_c^T \tilde{Z}_i U_t \in \mathbb{R}^{k_c\times k_t}$.
	α	Nonseparability of the planted truth in the controlled evidence ($\alpha = 0$: exactly separable).
	$\otimes$	Kronecker product; $\mathrm{vec}(AZB) = (B^T \otimes A)\mathrm{vec}(Z)$.
	$O(k_c)\times O(k_t)$	Product gauge of the separable class (Appendix S13.4).
Symmetry and loss	G_{noise}	Noise isometry group preserving Σ^{-1} (Definition S13.1).
	$\mathcal{U}(k)$	Latent gauge group of basis rotations (Definition S13.2).

Category	Symbol	Meaning
	M	Weight of a quadratic residual loss $\mathcal{L}_M$; $M = \kappa\Sigma^{-1}$ under noise-isometry invariance.
	κ	Positive scale of the invariant loss (Proposition 3.3).
	$\mathcal{F}$	Admissible reconstruction class; a method is the pair $(\Sigma^{-1}, \mathcal{F})$.
	B	Whitened reconstruction map $\tilde{D}\tilde{E}$ of an untied pair (Appendix S13.2).
Representation geometry	ϑ	Physical latent variables of a forward model $x = g(\vartheta) + n$, $\vartheta \in \mathbb{R}^m$.
	m	Number of physical latent variables.
	$g,\ J_g$	Forward model and its Jacobian $\partial g/\partial\vartheta \in \mathbb{R}^{d\times m}$.
	$I(\vartheta)$	Fisher information $J_g^T\Sigma^{-1}J_g$, the pulled-back noise metric (Proposition 2.4).
	$h,\ g_{\text{out}},\ J_h$	A representation, its output map to observation space, and the Jacobian $\partial y/\partial h$.
	$S,\ T_S$	Excited latent coordinates and the excited tangent span range$(\Sigma^{-1/2}A_{\cdot S})$ in whitened coordinates.
	R_t	Second moment BKB^T of the whitened signal displacement over the training distribution, $B = \Sigma^{-1/2}A_{\cdot S}$, $K = \mathbb{E}[\delta_S\delta_S^T]$.
	Π	Rank-k orthogonal projector in whitened coordinates.

S20 Configuration inventory

S21 Configuration Inventory

The operating point of every stage, and every run-level override, is listed in Table S20. The complete mechanically extracted inventory of the frozen configuration files (363 rows from 25 files, experiment P1-E4b) is not reproduced here: it is a separately built supplementary document, `supplement_configuration.pdf`, generated from the registered file `transfer_paper/data/paper1/P1-E4b/hyperparameters.csv` by `figures_src/make_paper1_evidence.py` and archived with the paper evidence. The inventory records what the configuration files say; which configuration a run used is recorded by the `config_hash` in that run's manifest.

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